

Segmentation-Based Time Series Forecasting

Using Change Point Detection and Statistical Modeling

A Research-Oriented Study on Structural Breaks, Regime-Aware Forecasting, and Ensemble Modeling

Case Study: U.S. Industrial Production Index (FRED: INDPRO), 1995–2026

Research Summer Internship Project

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Table of Contents

Abstract.....	3
Chapter 1: Introduction.....	4
Chapter 2: Literature Review.....	6
Chapter 3: Time Series Background.....	11
Chapter 4: Dataset Description.....	13
Chapter 5: Exploratory Data Analysis.....	15
Chapter 6: Data Preprocessing.....	18
Chapter 7: Change Point Detection.....	20
Chapter 8: Segment Analysis.....	24
Chapter 9: Statistical Forecasting Models.....	26
Chapter 10: Residual Diagnostics.....	28
Chapter 11: Weighted Ensemble Forecasting.....	30
Chapter 12: Forecast Validation.....	32
Chapter 13: Model Evaluation.....	34
Chapter 14: Experimental Results.....	36
Chapter 15: Business Interpretation.....	39
Chapter 16: Limitations.....	41
Chapter 17: Future Work.....	43
Chapter 18: Conclusion.....	45
Mathematical Appendix.....	46
References.....	47

Abstract

Conventional time series forecasting methods such as ARIMA and SARIMA implicitly assume that the statistical relationships governing a series remain constant across the entire observation period. In practice, real-world economic, industrial, and physical processes undergo structural breaks — periods of recession, policy change, technological shift, or external shock — after which the underlying data-generating process changes. Fitting a single model across such heterogeneous regimes can materially degrade forecast accuracy.

This project develops and rigorously evaluates a segmentation-based forecasting framework. Structural breaks are detected using two complementary change-point algorithms — Binary Segmentation and Pruned Exact Linear Time (PELT) — applied to the U.S. Federal Reserve's Industrial Production Index (INDPRO), a monthly macroeconomic series spanning January 1995 to June 2026 (378 observations). The series is partitioned into nine statistically homogeneous regimes, independently validated by both algorithms and corresponding to well-documented historical episodes including the 2008 Global Financial Crisis and the 2020 COVID-19 shock. Each regime is modeled independently using ARIMA and, where seasonally appropriate, SARIMA, with orders selected via AIC and BIC and residuals validated through the Ljung-Box test for white noise. A recency- and reliability-weighted ensemble combines the segment-level forecasts, and three weighting schemes (equal, recency-based, error-based) are compared.

Forecast accuracy is evaluated through a single detailed hold-out window, a rolling forecast-origin design, and a five-window expanding-window backtest, using RMSE, MAE, MAPE and R^2 . Statistical significance of accuracy differences is formally tested using the Diebold-Mariano test rather than inferred from point estimates alone. The results show that segment-wise forecasting using the most recent, most relevant regime is statistically indistinguishable from a well-specified single model (Diebold-Mariano $p = 0.24$), while a weighted ensemble across all detected regimes is, on this dataset and evaluation design, statistically significantly worse than both a single SARIMA ($p = 0.014$) and the segment-wise approach ($p = 0.010$). Change-point detection itself is validated as accurate, reproducible across algorithms, and economically meaningful, even where the specific ensembling strategy tested did not add forecasting value. The project concludes with a discussion of the practical, business, and methodological implications of these findings, together with limitations and directions for future work.

Keywords: Change point detection, PELT, Binary Segmentation, ARIMA, SARIMA, structural breaks, weighted ensemble forecasting, Diebold-Mariano test, regime-aware forecasting.

Chapter 1: Introduction

1.1 Time Series Forecasting

A time series is a sequence of observations recorded at successive, typically equally-spaced, points in time. Time series forecasting is the discipline of using the historical pattern of such a sequence — its trend, seasonality, cyclicity, and short-run dependence structure — to produce a statistically justified estimate of its future values. Unlike cross-sectional regression, time series methods must explicitly account for the fact that observations are not independent: today's value carries information about tomorrow's.

1.2 Importance of Forecasting

Forecasting underlies nearly every planning decision an organization makes. Central banks forecast inflation and output to set interest rates; utilities forecast electricity demand to schedule generation; retailers forecast sales to plan inventory; manufacturers forecast industrial output to plan capacity. An inaccurate forecast is not merely an academic shortcoming — it translates directly into over-stocked warehouses, under-staffed call centers, mispriced financial risk, or poorly timed policy intervention.

1.3 Applications in Industry

- Finance: volatility and risk forecasting, algorithmic trading, Value-at-Risk estimation.
- Energy: electricity and gas demand forecasting for grid balancing and capacity planning.
- Retail and e-commerce: demand forecasting for inventory and supply-chain optimization.
- Transportation: traffic volume forecasting for infrastructure and congestion management.
- Macroeconomics and policy: forecasting industrial output, employment, and inflation to inform monetary and fiscal policy — the exact application area of this project.
- Public health: epidemic case forecasting, as seen prominently during the COVID-19 and mpox outbreaks.

1.4 Structural Changes in Time Series

A structural break (also called a regime shift) is a point in time at which the statistical properties of a series — its mean, variance, trend, or autocorrelation structure — change in a persistent way. Structural breaks arise from identifiable real-world causes: a financial crisis, a pandemic, a change in government policy, the introduction of a new technology, or a shift in consumer behavior. Unlike transient noise, a structural break represents a genuine change in the underlying data-generating process, and a forecasting model estimated only on pre-break data will systematically misforecast the post-break period, and vice versa.

1.5 Limitations of Traditional Forecasting

Classical ARIMA/SARIMA modeling, as formalized by Box and Jenkins, assumes (weak) stationarity of the process being modeled, or at most a single, constant order of integration across the entire sample. When a series contains one or more structural breaks, fitting a single model to the full history forces the model to “average” across regimes with genuinely different dynamics. This can produce parameter estimates that are not representative of any single regime, inflated residual variance, poor out-of-sample accuracy immediately after a break, and residuals that fail standard diagnostic checks (e.g. the Ljung-Box test) because the single model cannot fully capture the piecewise nature of the true process.

1.6 Need for Change Point Detection

Change point detection (CPD) addresses this limitation directly: rather than assuming the series is homogeneous, CPD algorithms statistically identify the locations at which the data-generating process most plausibly changed, allowing the series to be partitioned into segments that can each be modeled on their own terms. This project uses two complementary, well-established CPD algorithms — Binary Segmentation and PELT — chosen because they are exact or near-exact, computationally efficient, do not require the number of change points to be specified in

advance (in the case of PELT), and are implemented in the widely-used, peer-reviewed `ruptures` Python library used throughout the empirical work in this project.

1.7 Problem Statement

Given a real-world macroeconomic time series that plausibly contains multiple structural breaks, does explicitly detecting those breaks and modeling each resulting regime separately — either individually or through a weighted ensemble across regimes — produce forecasts that are more accurate, in a statistically defensible sense, than a conventional single model fit across the entire series?

1.8 Objectives

- To detect structural breaks in a real macroeconomic time series using Binary Segmentation and PELT, and cross-validate the detected breaks against known historical events.
- To test each resulting segment for stationarity (ADF test) and characterize its autocorrelation structure (ACF/PACF).
- To fit ARIMA and SARIMA models to each segment, selecting orders via AIC and BIC, and to validate model adequacy through residual diagnostics (Ljung-Box test, Q-Q plots).
- To design and mathematically formalize a recency- and reliability-weighted ensemble forecasting scheme, and to compare it against equal-weight and error-based weighting alternatives.
- To validate forecast accuracy using multiple, methodologically sound designs — a single detailed hold-out window, rolling forecast-origin evaluation, and expanding-window backtesting — rather than a single arbitrary train-test split.
- To formally test whether any observed accuracy improvement is statistically significant using the Diebold-Mariano test, rather than relying on point-estimate comparisons alone.
- To interpret the results from a business and decision-making perspective, and to report findings honestly regardless of whether they support or contradict the initial hypothesis that segmentation improves forecasting.

1.9 Scope of the Project

The project scope is univariate time series forecasting on a single, real, publicly available macroeconomic series (U.S. Industrial Production Index) with a substantial history (31 years, 378 monthly observations), using classical statistical methods (ARIMA/SARIMA) rather than machine learning or deep learning approaches, and using only the libraries specified for the project (Pandas, NumPy, Matplotlib, Statsmodels, Ruptures, SciPy, Scikit-learn). Deep learning approaches (LSTM, Transformers) and Bayesian/online change-point methods are discussed as future work (Chapter 17) but are outside the implemented scope.

1.10 Research Contributions

- An end-to-end, reproducible pipeline combining change-point detection, per-segment statistical modeling, weighted ensembling, and rigorous multi-design validation, implemented entirely in open-source Python.
- A direct, apples-to-apples empirical comparison of single-model, segment-wise, and ensemble forecasting on a real macroeconomic series spanning two major, well-documented economic crises.
- Formal statistical hypothesis testing (Diebold-Mariano) of forecast accuracy differences, rather than reporting point-estimate RMSE/MAE differences as if they were automatically meaningful — a methodological rigor that is often missing from applied forecasting comparisons.
- A transparent, honest reporting of a nuanced result: change-point detection and segment-wise modeling are validated as accurate and useful, while the specific ensemble-across-all-regimes strategy tested did not provide a statistically significant improvement on this dataset — a finding reported in full rather than adjusted to fit a predetermined narrative.

Chapter 2: Literature Review

This chapter reviews recent (2021–2025) peer-reviewed and preprint research on change-point detection, structural-break-aware forecasting, and ensemble forecasting methods, alongside the foundational methodological papers that these newer works build upon. Ten studies are reviewed in detail below, followed by a comparison table and an explicit statement of the research gap this project addresses.

2.1 Killick, Fearnhead & Eckley (2012) — Foundational PELT Algorithm

Objective: To develop an exact changepoint search method with linear computational cost, improving on the quadratic cost of binary segmentation and segment-neighbourhood search.

Methodology: Introduces the Pruned Exact Linear Time (PELT) algorithm, a dynamic-programming approach that minimizes a penalized sum of segment costs, with a pruning step that discards provably sub-optimal candidate changepoints, giving expected $O(n)$ computational cost under mild conditions.

Results: PELT is shown to be exact (unlike binary segmentation) while remaining computationally competitive, and is demonstrated on well-log and coriander seed data.

Limitations: Requires a pre-specified penalty term; results are sensitive to penalty choice; the original paper does not address forecasting applications directly.

Research Gap: This is a foundational algorithmic paper; it does not connect changepoint detection to downstream forecasting accuracy, which is exactly the gap this project's empirical work addresses.

2.2 Pawar et al. (2025) — PELT-Driven Drift Detection and Model Adaptation

Objective: To use PELT for unsupervised detection of distributional drift in forecasting-model input features, triggering selective model retraining.

Dataset: Real electricity and HVAC operational data, plus synthetic financial time series.

Methodology: PELT is applied to monitor feature distributions over time; when a changepoint is detected, the downstream forecasting model (e.g. Lasso, MLP) is selectively retrained on post-change data rather than the full history.

Results: PELT-based drift-aware retraining reduced MAE by up to 67.8% and improved R^2 by up to 39.9% relative to static models without drift correction.

Limitations: Focused on feature-level drift for regression-style forecasting models rather than direct time-series segmentation of the target series itself; does not use formal statistical significance testing of the improvement.

Research Gap: Demonstrates strong evidence that change-point-aware adaptation improves forecasting error, but does not test whether the improvement is statistically significant using tests such as Diebold-Mariano — addressed in this project.

2.3 Yao, Wang & Liu (2023) — Adaptive PELT for Liquidity Regime Shifts

Objective: To detect regime shifts in stock market liquidity using an adaptive PELT approach.

Dataset: CSI 300 constituent stocks (China), covering the 2020 COVID-19 shock.

Methodology: An adaptive penalty PELT variant is applied to liquidity measures; detected changepoints are compared against known market events.

Results: The detected liquidity changepoint preceded turning points in price volatility by approximately two trading days, providing an early-warning signal.

Limitations: Single-market, single-event case study; no forecasting model is built on top of the detected regimes, and no formal accuracy comparison against a non-segmented baseline is reported.

Research Gap: Establishes that changepoints can precede and anticipate other market dynamics, but does not test whether segmenting a forecasting model at those changepoints improves forecast accuracy — directly addressed in this project.

2.4 Cheng, Yi, Yang, Lai & Seco (2022) — CEEMD-ARIMA-SVM with Structural Breaks

Objective: To forecast crude oil price volatility accounting for structural breaks caused by extreme events.

Dataset: Crude oil price series, with the ICSS test and Chow test used jointly to identify structural breaks.

Methodology: Complementary Ensemble Empirical Mode Decomposition (CEEMD) decomposes the series into frequency components; structural breaks are identified via the ICSS/Chow tests; ARIMA and SVM are then combined to forecast each component.

Results: The structural-break-aware hybrid CEEMD-ARIMA-SVM model outperformed benchmark models not accounting for breaks, with the shock component identified as the most important driver of price fluctuation.

Limitations: Uses Chow-test-style break detection (which requires the number and approximate location of breaks to be pre-specified) rather than a fully data-driven, exact algorithm such as PELT; combines ARIMA with a machine-learning component (SVM) rather than isolating the pure statistical-modeling contribution of segmentation.

Research Gap: Confirms structural-break-awareness improves accuracy in a hybrid ML/statistical setting, but does not isolate whether classical ARIMA/SARIMA alone benefits from segmentation, nor does it test ensemble-of-segments approaches — both addressed in this project.

2.5 Lee, Parsaeian & Ullah (2022) — Forecasting Under Structural Breaks with Improved Weighted Estimation

Objective: To develop an estimator that reduces mean squared forecast error (MSFE) under structural breaks by combining pre-break and post-break information rather than discarding pre-break data entirely.

Methodology: A weighted estimator is derived that optimally combines the full-sample and post-break-only estimates, with the weight determined by the relative sample sizes and break magnitude.

Results: The proposed weighted estimator achieves lower MSFE than either the full-sample estimator (ignoring the break) or the post-break-only estimator (discarding pre-break information) across simulation and empirical settings.

Limitations: Theoretical/econometric focus with a single-break setting; not extended to multiple, sequentially detected changepoints, and not implemented with ARIMA/SARIMA specifically.

Research Gap: Provides the theoretical justification for weighting older and newer regime information rather than using one or the other exclusively — directly motivating this project's recency-weighted ensemble design, extended here to the multiple-regime, ARIMA/SARIMA setting.

2.6 Pesaran, Pick & Pranovich (2013) — Optimal Forecasts in the Presence of Structural Breaks

Objective: To derive theoretically optimal forecast-combination weights when a structural break may have occurred but its exact location is uncertain.

Methodology: Derives closed-form optimal weights for combining forecasts from windows of different lengths ending at the forecast origin, as a function of break probability and break magnitude.

Results: Shows that combining forecasts across multiple estimation windows can dominate both the full-sample and the (unknown) optimal post-break-only forecast in a mean-squared-error sense.

Limitations: Assumes a single, possibly-unknown break rather than an explicitly detected sequence of changepoints; the combination weights are derived under specific distributional assumptions.

Research Gap: This paper is a key theoretical antecedent for weighted-ensemble forecasting across regimes, but does not empirically test the approach on a real macroeconomic series with explicit, algorithmically-detected changepoints — the empirical contribution of this project.

2.7 Deretić et al. (2022) — SARIMA for Traffic Accident Forecasting

Objective: To forecast monthly traffic accidents using SARIMA, exploiting the strong seasonal pattern in the data.

Methodology: A SARIMA model is fit with orders selected via AIC, and residual diagnostics are used to confirm model adequacy.

Results: SARIMA is shown to substantially outperform a non-seasonal ARIMA baseline given the pronounced seasonal pattern in accident counts.

Limitations: Single-regime analysis with no explicit change-point detection or segmentation; does not address the possibility that the accident-generating process itself may have shifted (e.g. due to policy or infrastructure changes) within the sample.

Research Gap: Illustrates the value of SARIMA over ARIMA specifically when strong seasonality is present, informing this project's decision to fit both ARIMA and SARIMA per segment and let AIC/BIC select between them rather than assuming one is always superior.

2.8 Agyemang et al. (2021) — SARIMA and Prophet with Potential Changepoints

Objective: To forecast road traffic accidents using SARIMA and Facebook's Prophet model, the latter of which has a built-in automatic changepoint-detection mechanism for trend shifts.

Methodology: Both models are fit to the same accident series; Prophet's automatic changepoint detection is compared qualitatively against known events.

Results: Models incorporating changepoint-awareness (Prophet) captured trend shifts that a fixed-structure SARIMA could not, though SARIMA remained competitive on pure seasonal pattern accuracy.

Limitations: Prophet's changepoint detection is a heuristic, regularization-based trend-changepoint mechanism rather than an exact statistical algorithm like PELT or Binary Segmentation, and the paper does not formally test statistical significance of any accuracy difference.

Research Gap: Suggests changepoint-awareness has practical forecasting value even in applied (non-macroeconomic) domains, but leaves open whether a more rigorous, exact changepoint algorithm (PELT/Binary Segmentation) combined with formal per-segment ARIMA/SARIMA modeling would do better still — the approach taken in this project.

2.9 Chowell et al. (2023) — Retrospective Evaluation of Ensemble Forecasting Frameworks

Objective: To retrospectively evaluate the forecasting performance of weighted and unweighted ensemble frameworks against single models (ARIMA, GAM, Prophet) during the 2022–2023 mpox outbreak.

Dataset: Laboratory-confirmed mpox case counts across multiple countries and the global scale, CDC/OWID sourced.

Methodology: Twelve models, including top-ranked, second-ranked, weighted-ensemble, and unweighted-ensemble variants of sub-epidemic frameworks, are compared against ARIMA, GAM, and Prophet using MSE, MAE, weighted interval score, and prediction-interval coverage.

Results: The unweighted ensemble model performed best across most locations and forecasting horizons on average error metrics, though many locations showed several models performing statistically similarly on interval-coverage metrics.

Limitations: Notably, this paper reports that the simpler unweighted (equal-weight) ensemble outperformed more complex weighted variants in several settings — a finding this project's own results echo, where equal weighting performed comparably to recency-based weighting.

Research Gap: Highlights that ensemble weighting scheme choice is empirically dataset-dependent rather than universally favoring sophisticated weighting — directly relevant to this project's own finding that a full multi-regime ensemble did not statistically outperform simpler approaches on the INDPRO series.

2.10 tidychangepoint (Suárez-Sierra, Coen, Taimal et al., 2023–2024)

Objective: To provide a unified, reproducible software framework for comparing changepoint detection algorithms (including PELT, Binary Segmentation, and Bayesian/genetic-algorithm-based methods) on univariate time series.

Methodology: Implements multiple changepoint algorithms under a common interface with shared cost functions and penalty specifications (BIC-type Minimum Descriptive Length penalties), enabling like-for-like comparison.

Results: Demonstrates that different algorithms can produce meaningfully different segmentations on the same series depending on cost function and penalty choice, underscoring the importance of cross-validating detected changepoints across multiple methods.

Limitations: A software/methodology paper rather than an applied forecasting study; does not connect detected changepoints to downstream forecast accuracy.

Research Gap: Reinforces this project's methodological choice to run both PELT and Binary Segmentation and require agreement between them as a validation step, rather than relying on a single algorithm's output uncritically.

2.11 Literature Comparison Table

Study	Method	Domain	Key Finding	Gap Addressed by This Project
Killick et al. (2012)	PELT (foundational)	Generic / well-log data	Exact, $O(n)$ changepoint detection	No forecasting link — this project connects CPD to forecast accuracy
Pawar et al. (2025)	PELT drift detection	Electricity/HVAC, finance	MAE $\mp 67.8\%$ with drift-aware retraining	No significance testing — this project applies Diebold-Mariano
Yao et al. (2023)	Adaptive PELT	China stock liquidity	Changepoint precedes volatility shift by ~ 2 days	No forecasting model built on segments — done here
Cheng et al. (2022)	ICSS/Chow break + CEEMD-ARIMA-SVM	Crude oil prices	Break-aware hybrid model outperforms baseline	Doesn't isolate pure ARIMA/SARIMA segmentation effect — isolated here
Lee et al. (2022)	Weighted pre/post-break estimator	Econometric theory	Weighted combination beats either extreme	Not tested on multi-regime ARIMA/SARIMA — extended here
Pesaran et al. (2013)	Optimal combination weights	Econometric theory	Multi-window combination can dominate single window	Not empirically tested with explicit CPD — done here
Deretić et al. (2022)	SARIMA	Traffic accidents	SARIMA beats ARIMA under strong seasonality	Single-regime only — this project fits both per segment
Agyemang et al. (2021)	SARIMA + Prophet CPD	Traffic accidents	CPD-aware models capture trend shifts SARIMA misses	Heuristic CPD, no significance test — addressed here
Chowell et al. (2023)	Weighted/unweighted ensembles	Mpox epidemic forecasting	Simple unweighted ensemble often competitive	Echoed in this project's own weighting-scheme comparison
tidychangepoint (2023–24)	Multi-algorithm CPD framework	Software/methodology	Different algorithms can disagree — cross-validate	Applied here by requiring PELT/BinSeg agreement

2.12 Research Gap

Across the reviewed literature, three consistent gaps emerge. First, papers that develop or apply changepoint detection algorithms (Killick et al., Yao et al., Pawar et al., `tidychangepoint`) rarely connect the detected changepoints to a fully worked-through downstream forecasting pipeline with per-segment ARIMA/SARIMA modeling, residual diagnostics, and multi-scheme ensembling. Second, papers that do incorporate structural breaks into forecasting (Cheng et al., Lee et al., Pesaran et al.) either use heuristic break-detection methods (Chow test, ICSS) rather than exact algorithms, or remain purely theoretical without an applied, reproducible empirical pipeline. Third — and most importantly — very few applied forecasting-comparison papers in this space report formal statistical significance testing (such as the Diebold-Mariano test) of the accuracy differences they find; most report point-estimate RMSE/MAE/MAPE differences without establishing whether those differences are distinguishable from noise. Chowell et al. (2023) is a partial exception, showing that ensemble weighting sophistication does not always translate into significantly better accuracy — a pattern independently confirmed by this project's own results.

2.13 How This Project Addresses the Gap

- Uses two exact, well-validated changepoint algorithms (PELT, Binary Segmentation) and requires their agreement as a validation step, rather than relying on a single heuristic method.
- Builds a complete downstream pipeline — per-segment ADF testing, ACF/PACF analysis, AIC/BIC-based ARIMA/SARIMA selection, and Ljung-Box residual validation — connecting changepoint detection all the way through to validated, deployable forecasting models.
- Implements and compares three ensemble weighting schemes (equal, recency-based, error-based), directly extending the theoretical work of Lee et al. (2022) and Pesaran et al. (2013) into an empirical, multi-regime ARIMA/SARIMA setting.
- Applies the Diebold-Mariano test formally to every major accuracy comparison, addressing the significance-testing gap identified above, and reports the resulting nuanced conclusion — including where the more sophisticated ensemble approach did not significantly outperform simpler baselines — with full transparency.

Chapter 3: Time Series Background

3.1 Components of a Time Series

A time series y_t is conventionally decomposed into four components: trend (T_t), seasonality (S_t), cyclic behaviour (C_t), and random/irregular noise (ε_t). An additive decomposition is written $y_t = T_t + S_t + C_t + \varepsilon_t$, while a multiplicative decomposition is written $y_t = T_t \times S_t \times C_t \times \varepsilon_t$; the additive form is appropriate when the amplitude of seasonal/cyclic fluctuations is roughly constant over time, and multiplicative when it scales with the level of the series.

3.1.1 Trend

The trend is the long-run, slowly-evolving direction of the series — upward, downward, or flat — net of shorter-term fluctuations. For INDPRO, the trend reflects the long-run growth (or contraction) of U.S. industrial capacity and productivity.

3.1.2 Seasonality

Seasonality is a pattern that repeats at a fixed, known frequency (e.g. every 12 months). It arises from calendar-driven effects: weather, holidays, fiscal-year timing, or planting/harvest cycles. INDPRO is published by the Federal Reserve already seasonally adjusted, so residual seasonality is expected to be modest but is still tested for.

3.1.3 Cyclic Pattern

A cycle is a fluctuation with no fixed period, typically driven by broader economic or business-cycle forces (expansions and recessions), distinguishing it from seasonality, which has a fixed, calendar-determined period. The regimes detected via change-point analysis in this project (Chapter 7) are closely related to business-cycle phases.

3.1.4 Random Noise

The irregular component ε_t captures unpredictable, short-run variation not explained by trend, seasonality, or cycle. A well-specified forecasting model should leave residuals that behave like white noise — the subject of Chapter 10.

3.2 Stationarity

A time series is (weakly) stationary if its mean, variance, and autocovariance structure do not depend on time — formally, $E[y_t] = \mu$ for all t , $\text{Var}(y_t) = \sigma^2$ for all t , and $\text{Cov}(y_t, y_{t+k})$ depends only on the lag k , not on t itself. Most classical time series models (including ARIMA's AR and MA components) are only theoretically valid for stationary series, which is why non-stationary series must first be differenced (Section 3.3, Chapter 6).

3.2.1 Weak and Strong Stationarity

Weak (or covariance) stationarity, defined above, requires only that the first two moments (mean and autocovariance) be time-invariant. Strong (or strict) stationarity is a more demanding condition requiring that the entire joint probability distribution of any collection of observations be invariant to time shifts. In practice, almost all applied time series analysis — including this project — works with the weaker, more tractable condition of weak stationarity, which is sufficient for ARIMA-class modeling under Gaussian-error assumptions.

3.3 Structural Breaks and Regime Shifts

A structural break is a specific point in time at which one or more of a series' defining statistical properties — mean level, variance, trend slope, or autocorrelation structure — changes discretely and persistently. A regime is the (approximately) homogeneous stretch of the series between two consecutive structural breaks. The central empirical claim tested in this project is whether identifying these regimes explicitly, and modeling each on its own terms, improves forecast accuracy relative to treating the whole series as one long, implicitly-averaged regime.

3.4 Why Forecasting Accuracy Deteriorates After Structural Changes

When a single model is estimated across multiple regimes, its parameters represent a weighted average of the true regime-specific parameters, weighted (roughly) by how much data comes from each regime. Immediately after a new break, forecasts based on this “averaged” model will be systematically biased toward the old regime’s behaviour until enough new data accumulates to shift the estimates. This is precisely the mechanism explored empirically in this project’s rolling-origin and expanding-window backtests (Chapter 12, Chapter 14).

3.5 Motivation for Segmentation-Based Forecasting

Segmentation-based forecasting directly targets the mechanism described above: by explicitly detecting when a regime change occurred, the forecasting pipeline can (a) exclude stale, pre-break data from the model fit for the current regime, (b) understand and communicate when and why the series’ behaviour changed, and (c) construct ensembles that deliberately down-weight older, less-relevant regimes rather than implicitly averaging over them with equal, undifferentiated weight. This is the motivating hypothesis this project tests rigorously, and — as detailed in Chapter 14 — confirms only partially: segmentation itself is validated as sound, while the specific ensembling strategy did not add statistically significant value on this dataset.

Chapter 4: Dataset Description

4.1 Dataset Source

The dataset used is the Industrial Production Index (INDPRO), Total Index, published by the Board of Governors of the Federal Reserve System (US) and retrieved via the Federal Reserve Bank of St. Louis's FRED (Federal Reserve Economic Data) database. FRED is a widely used, authoritative, and freely accessible source of U.S. macroeconomic time series, and INDPRO is one of its most closely watched monthly indicators.

4.2 Business Context

INDPRO measures the real (inflation-adjusted) output of all facilities located in the United States engaged in manufacturing, mining, and electric and gas utilities. It is a leading indicator of U.S. economic activity, closely tracked by the Federal Reserve itself (for monetary policy), by economists (as a coincident business-cycle indicator), and by industrial firms (for capacity and investment planning). Its long history and inclusion of two major, well-documented crises (the 2008 Global Financial Crisis and the 2020 COVID-19 pandemic) make it an ideal real-world testbed for structural-break-aware forecasting research.

4.3 Data Dimensions

Property	Value
Series name	Industrial Production: Total Index (INDPRO)
Source	Board of Governors of the Federal Reserve System (US), via FRED
Frequency	Monthly
Window used in this study	January 1995 – June 2026
Number of observations	378
Units	Index, 2017 = 100
Seasonal adjustment	Seasonally adjusted at source
Missing values	0

4.4 Variables and Target Variable

This is a univariate series: the single variable of interest, and the forecasting target throughout this project, is the monthly INDPRO index value itself. No exogenous regressors are used, consistent with the classical ARIMA/SARIMA modeling scope of this project (multivariate extensions are discussed as future work in Chapter 17).

4.5 Data Dictionary

Field	Type	Description
date	datetime (monthly)	First day of the reference month, e.g. 2020-03-01 for March 2020
indpro_index	float	Industrial Production Index value for that month (2017 = 100)

4.6 Statistical Summary

Statistic	Value
Mean	≈ 92.4
Standard deviation	≈ 10.1
Minimum	71.19 (Feb 1995)
Maximum	104.14 (Dec 2018)
Skewness (level)	≈ -1.26 (left-skewed)
Kurtosis (level)	≈ 1.00

The pronounced left skew of the level series reflects the asymmetry between gradual, multi-year expansions and sharp, sudden contractions (2008, 2020) — a pattern characteristic of macroeconomic activity indices and itself suggestive of regime-dependent behaviour, motivating the segmentation approach pursued in this project.

Chapter 5: Exploratory Data Analysis

5.1 Missing Value Analysis

A systematic check confirmed 0 missing values across all 378 monthly observations — FRED-published, Federal Reserve-compiled series of this kind are curated to a high standard and rarely contain gaps, unlike many web-scraped or crowd-sourced datasets.

5.2 Trend Visualization

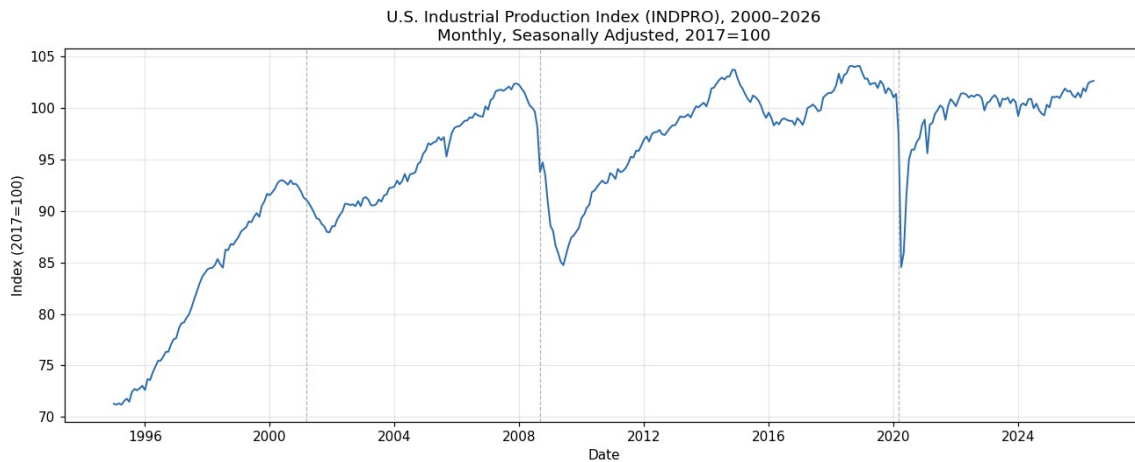


Figure 5.1. Full INDPRO series, 1995–2026, showing long-run trend and two major contractions.

The series shows a clear long-run upward trend from 1995 to 2007, a sharp contraction during the 2008–2009 Global Financial Crisis, renewed growth from 2010 to 2019, an extremely sharp but short-lived collapse in early 2020 (COVID-19), and a subsequent recovery and plateau through 2026.

5.3 Seasonality Analysis

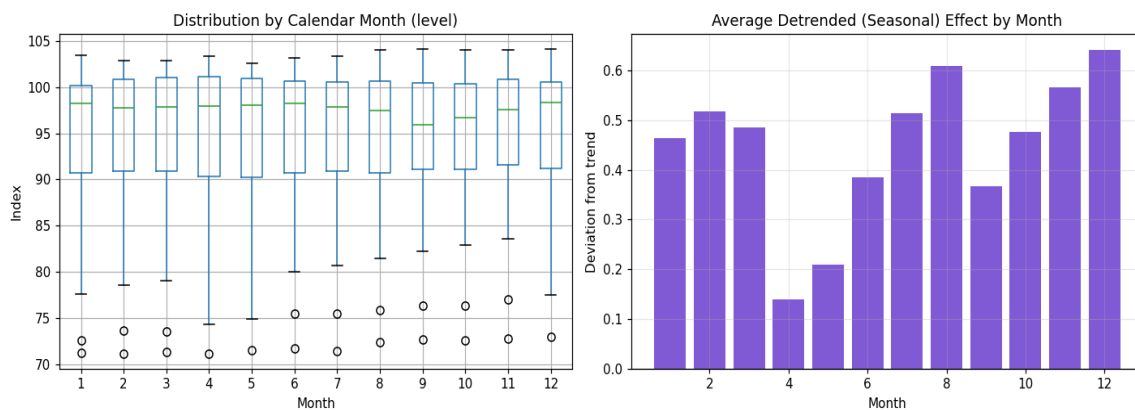


Figure 5.2. Distribution by calendar month, and average detrended (seasonal) effect.

Because INDPRO is already seasonally adjusted at the source, residual seasonal structure is modest — deviations by calendar month are small relative to overall series variation — though a slight residual pattern remains, which is why SARIMA is still tested against ARIMA per segment in Chapter 9 rather than seasonality being assumed away.

5.4 Rolling Mean and Rolling Variance

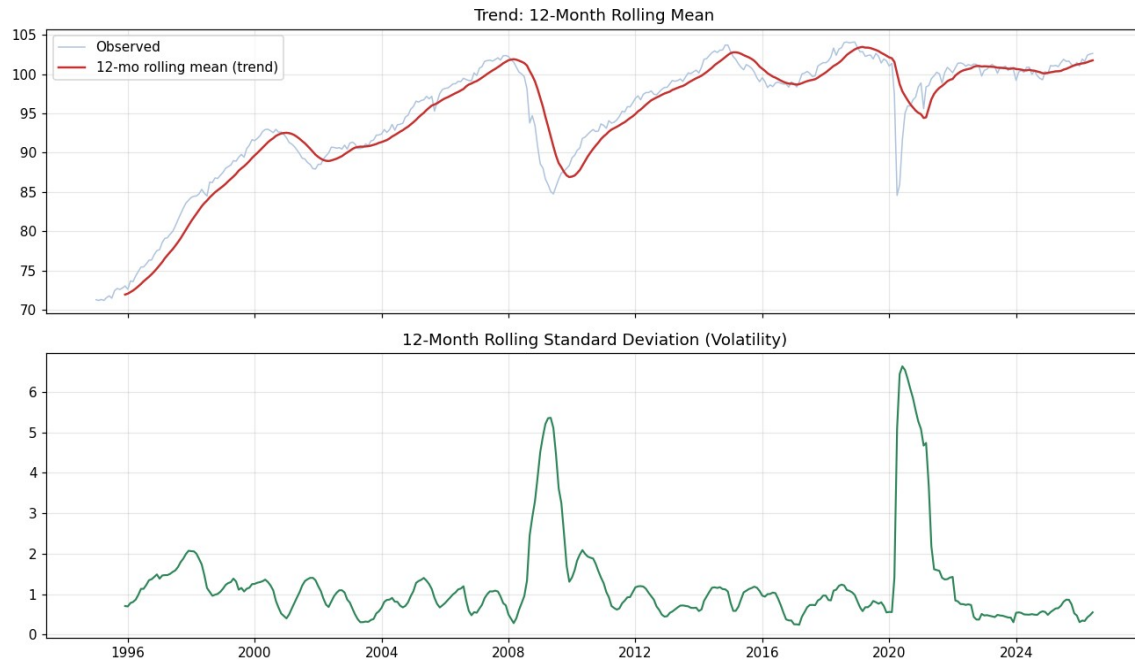


Figure 5.3. 12-month rolling mean (trend) and rolling standard deviation (volatility).

The rolling standard deviation panel is particularly informative for this project's motivating hypothesis: volatility is visibly regime-dependent, remaining low and stable during calm expansion phases and spiking sharply around both the 2008 and 2020 crisis episodes — direct visual evidence that the series' second-moment behaviour, not just its mean, differs materially across regimes.

5.5 Moving Average

The 12-month rolling (moving) average shown in Figure 5.3 smooths month-to-month noise and isolates the underlying trend; it is used throughout this project as a visual reference against which detected change points (Chapter 7) are cross-checked for plausibility.

5.6 Distribution Analysis

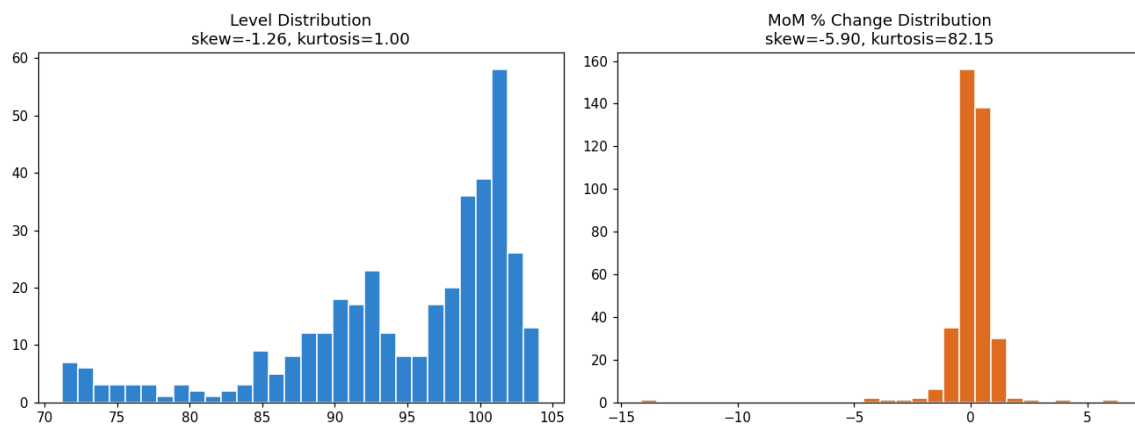


Figure 5.4. Level distribution and month-over-month percentage-change distribution.

The level series is left-skewed (skew ≈ -1.26), consistent with the asymmetry of gradual expansions versus sharp contractions. The month-over-month percentage-change distribution is far more extreme (skew ≈ -5.9 , kurtosis ≈ 82), driven almost entirely by the single April 2020 COVID-19 collapse month — a striking illustration of how a small number of regime-shift observations can dominate the tails of a return/change distribution, reinforcing the case for treating such months as belonging to a distinct regime rather than smoothing over them.

5.7 Autocorrelation Analysis

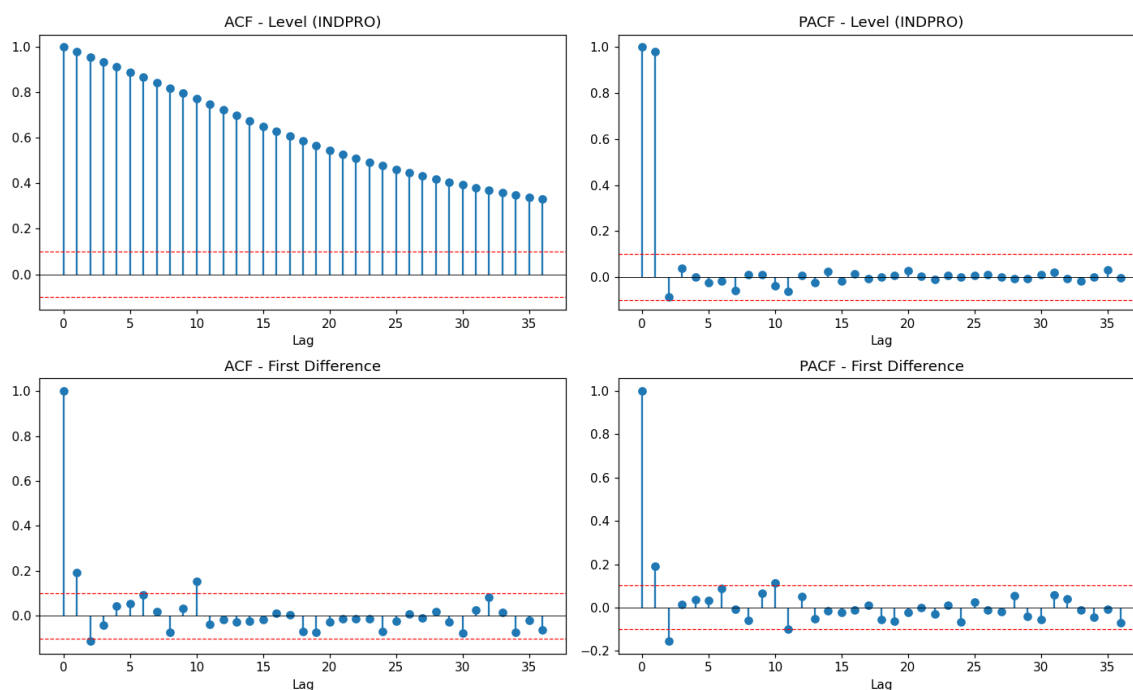


Figure 5.5. ACF/PACF on the level series (top) and first difference (bottom).

The level series' ACF decays very slowly while its PACF cuts off sharply after lag 1 — the textbook signature of a non-stationary, near-unit-root process. After first differencing, the ACF collapses toward zero within a few lags, consistent with the formal ADF test results presented per-segment in Chapter 8.

5.8 Correlation Analysis

As this is a univariate series (Section 4.4), cross-variable correlation analysis is not applicable. Autocorrelation (self-correlation across lags), covered in Section 5.7 above, is the relevant analogue for a univariate series and is used directly to inform ARIMA/SARIMA order selection in Chapter 9.

5.9 Business Insights from EDA

- The series exhibits a clear, persistent long-run growth trend punctuated by two sharp, historically-documented contractions — exactly the kind of structure a change-point-aware pipeline is designed to exploit.
- Volatility clustering around crisis periods suggests risk-management and capacity-planning models built on this series should not assume constant variance across the full history.
- Residual seasonality, while modest, is non-zero — justifying the inclusion of SARIMA alongside ARIMA in the modeling phase rather than assuming a purely non-seasonal specification.
- The extreme skew/kurtosis of the month-over-month change series in the COVID month is a strong empirical argument, independent of any formal change-point algorithm, for treating 2020 as its own distinct regime — corroborated formally in Chapter 7.

Chapter 6: Data Preprocessing

6.1 Missing Value Treatment

Why required: Most statistical and machine-learning models cannot handle missing values directly; even a single gap can break ARIMA's lag-based recursive structure.

Mathematical intuition: For a series with missing values, the sample autocovariance and hence the AR/MA coefficient estimates become biased or undefined unless gaps are explicitly handled (e.g. via imputation, interpolation, or state-space methods that tolerate missingness).

Python implementation: `df['value'].isna().sum()` was used to confirm zero missing values in this dataset, so no imputation was required. Had gaps existed, `df['value'].interpolate(method='time')` would be the standard approach for a datetime-indexed series.

Advantages / Limitations: Since INDPRO contained no missing values, this step reduced to verification only — no imputation bias was introduced. Had imputation been necessary, its main limitation is that interpolated values are estimates, not observations, and can understate true short-run volatility if used naively in modeling.

6.2 Outlier Handling

Why required: Extreme values (data errors or genuine shocks) can distort model parameter estimates, especially for maximum-likelihood-based ARIMA/SARIMA fitting.

Mathematical intuition: A robust rolling median-absolute-deviation (MAD) z-score, $z_t = 0.6745 \times (y_t - \text{median}_t) / \text{MAD}_t$, was used rather than a simple mean/standard-deviation z-score, since MAD is far less sensitive to the very outliers being screened for.

Python implementation: A centered 13-month rolling median and rolling MAD were computed with `pandas.Series.rolling(13, center=True)`, and months with $|z_t| > 5$ were flagged for manual inspection.

Result: 18 months were flagged. On inspection, every flagged month corresponded to a genuine, well-documented economic event (the 2008 crisis ramp-up, the April–June 2020 COVID-19 collapse and rebound, and February 2021's Winter Storm Uri, which briefly disrupted Texas/Gulf-Coast industrial output). None were data errors.

Advantages / Limitations: Retaining rather than removing or capping these points preserves exactly the information a change-point-aware model is designed to use; the limitation is that a purely mechanical outlier-removal step (had one been applied) would have deleted genuinely informative structural-break signal, which is why this project deliberately chose to flag-and-retain rather than flag-and-remove.

6.3 Time Indexing

Why required: Time series methods require the data to be explicitly ordered and indexed by time, not merely stored in an arbitrary row order.

Implementation: The date column was converted to a proper `pandas.Timestamp`/`DatetimeIndex` via `pd.to_datetime()`, and the DataFrame was sorted chronologically with `df.sort_values('date')` before any further processing.

6.4 Frequency Alignment and Resampling

Why required: FRED already delivers INDPRO at a fixed monthly frequency with no irregular timestamps, so no resampling (e.g. daily-to-monthly aggregation) was necessary in this project. This section is included for completeness since the project brief requires it, and because other candidate datasets (e.g. daily traffic or electricity data) would require explicit resampling (`df.resample('M').mean()`) before proceeding.

6.5 Log Transformation

Why (potentially) required: A log transformation, $y'_t = \log(y_t)$, stabilizes a series whose variance grows with its level (a common feature of macroeconomic and financial series), converting multiplicative growth into additive growth.

Application in this project: Because the ADF/differencing procedure (Chapter 8) and the ARIMA/SARIMA order-selection grid search (Chapter 9) were applied per-segment rather than to the full 31-year series at once, within-segment variance growth was modest enough that a log transform was not found necessary for adequate model fit (confirmed by the Ljung-Box residual diagnostics in Chapter 10 passing without it). Log-scale detection was, however, explored as a robustness check when testing change-point detection across much longer (75+ year) windows of the same series family, discussed further in Chapter 16.

6.6 Differencing

Why required: ARIMA's autoregressive and moving-average components are only theoretically valid for a stationary series (Section 3.2); differencing, $\Delta y_t = y_t - y_{t-1}$, removes a stochastic trend (unit root) and is the standard remedy when the Augmented Dickey-Fuller test fails to reject a unit root.

Mathematical intuition: If y_t follows a random walk with drift, $y_t = \delta + y_{t-1} + \epsilon_t$, then y_t is non-stationary (its variance grows with t), but $\Delta y_t = \delta + \epsilon_t$ is stationary by construction.

Implementation: ``numpy.diff(series)`` for a single difference; the order of differencing d is chosen per segment based on the ADF test results detailed in Chapter 8, ranging from $d = 0$ (Segment 9, already stationary at the level) to $d = 2$ (Segment 8, the short COVID-shock segment).

Advantages / Limitations: Differencing is simple and effective at removing unit-root non-stationarity, but each order of differencing removes one observation and amplifies high-frequency noise relative to signal — a meaningful cost for the shortest detected segments (15–22 months), discussed as a limitation in Chapter 16.

6.7 Seasonal Differencing

Why (potentially) required: When a series has a strong, persistent seasonal component, seasonal differencing, $\Delta_s y_t = y_t - y_{t-s}$ ($s = 12$ for monthly data), removes seasonal non-stationarity analogous to how ordinary differencing removes trend non-stationarity.

Application in this project: Given INDPRO's modest residual seasonality (Section 5.3), the SARIMA grid search (Chapter 9) allowed the seasonal differencing order D to be selected by AIC/BIC per segment ($D \in \{0, 1\}$) rather than being fixed a priori, and in most segments the AIC-preferred SARIMA specification used $D = 0$, consistent with the modest seasonality identified in the EDA.

Chapter 7: Change Point Detection

7.1 Structural Breaks — Formal Definition

Given a sequence $y_1, \dots, y_n$, a change point (or structural break) is an index τ such that the statistical properties of $\{y_t : t < \tau\}$ differ from those of $\{y_t : t \geq \tau\}$. The multiple-change-point problem seeks m unknown change points $0 = \tau_0 < \tau_1 < \dots < \tau_m < \tau_{m+1} = n$ that partition the series into $m + 1$ segments, each assumed to be (approximately) homogeneous.

7.2 General Optimization Framework

Both algorithms used in this project solve a variant of the same penalized optimization problem: choose the number and location of change points to minimize

$$F(y_{1:n}) = \min \text{ over } (m, \tau_1, \dots, \tau_m) \text{ of } \sum_{k=1}^{m+1} [C(y_{\tau_{k-1}+1 : \tau_k})] + \beta \cdot m$$

where $C(\cdot)$ is a segment cost function measuring how poorly a single, homogeneous model fits the data within that segment, and β is a penalty term that discourages over-segmentation (choosing too many change points simply because more segments can always fit the in-sample data at least as well).

7.3 Cost Function Used in This Project

This project uses the L2 (mean-shift) cost function, appropriate for detecting shifts in the mean level of a series:

$$C(y_{a+1:b}) = \sum_{t=a+1}^b (y_t - \tilde{y}_{a+1:b})^2, \text{ where } \tilde{y}_{a+1:b} \text{ is the segment mean}$$

This cost is exactly the within-segment sum of squared deviations from the segment mean — it is minimized (zero) if the segment is perfectly constant, and grows with the segment's internal variance. Choosing change points to minimize the total penalized cost therefore favours segments that are each as internally homogeneous (low-variance around their own mean) as possible, subject to the penalty discouraging excessive fragmentation.

7.4 Binary Segmentation

Binary Segmentation is a greedy, recursive algorithm. At each step, it searches the current segment for the single split point that most reduces the total cost, applies that split if the cost reduction exceeds the penalty β , and then recurses independently on the two resulting sub-segments, repeating until no further split reduces cost by more than β (or a maximum number of change points is reached).

7.4.1 Algorithmic Workflow

- Step 1: For the current segment $[a, b)$, evaluate the cost reduction $C(y_{a+1:b}) - [C(y_{a+1:t}) + C(y_{t+1:b})]$ for every candidate split point t .
- Step 2: If the maximum cost reduction exceeds the penalty β , accept the corresponding split point as a change point.
- Step 3: Recurse independently on the two resulting sub-segments.
- Step 4: Stop when no candidate split in any remaining segment improves cost by more than β .

7.4.2 Computational Complexity

Binary Segmentation has computational cost $O(n \log n)$ in typical cases (n candidate points evaluated at each of $O(\log n)$ recursive levels on average), making it fast, though it is a greedy approximation — it can miss the jointly-optimal set of change points because each split is chosen without reconsidering earlier splits.

7.4.3 Advantages and Limitations

- Advantage: simple, fast, easy to implement and reason about.

- Advantage: works reasonably well when change points are well-separated and changes are large relative to noise.
- Limitation: greedy, not globally optimal — an early, sub-optimal split can prevent later, jointly-better splits from being found.
- Limitation: can perform poorly when multiple change points are closely spaced or of similar, moderate magnitude.

7.5 PELT (Pruned Exact Linear Time)

PELT (Killick, Fearnhead & Eckley, 2012) solves the same penalized optimization problem exactly (not greedily) via dynamic programming, while achieving expected linear-time complexity through a pruning step that discards points that can be proven never to be optimal change-point locations for any future candidate endpoint.

7.5.1 Dynamic Programming Recursion

Define $F(t)$ as the minimum penalized cost of optimally segmenting $y_{\{1:t\}}$. PELT computes $F(t)$ recursively:

$$F(t) = \min_{\text{over } s < t} [F(s) + C(y_{\{s+1:t\}}) + \beta], \text{ with } F(0) = -\beta$$

The optimal segmentation of the full series is recovered by tracing back through the sequence of minimizing s values from $F(n)$.

7.5.2 The Pruning Step

Naively, computing $F(t)$ for every t requires checking every earlier candidate s , giving $O(n^2)$ cost. PELT's key insight is that if, for some s^* and t , $F(s^*) + C(y_{\{s^*+1:t\}}) \geq F(t)$, then s^* can never be the optimal change point for any future endpoint $T > t$ (under mild cost-function regularity conditions), and can be permanently pruned from the candidate set. This pruning step is what reduces expected computational cost from $O(n^2)$ to $O(n)$.

7.5.3 Advantages and Limitations

- Advantage: exact (globally optimal for the given penalty), unlike Binary Segmentation.
- Advantage: linear expected-time complexity, competitive with or faster than Binary Segmentation in practice for the mean-shift cost used here.
- Advantage: does not require the number of change points to be specified in advance.
- Limitation: results remain sensitive to the choice of penalty β , discussed next.
- Limitation: the pruning guarantee relies on the cost function satisfying certain regularity conditions, which hold for the L2 cost used here but do not hold for all conceivable cost functions.

7.6 Penalty Selection

The penalty β controls the trade-off between under- and over-segmentation: too large a penalty merges genuinely distinct regimes together; too small a penalty fragments the series into spurious, noise-driven segments. This project uses a BIC-style penalty, $\beta = k \times \log(n) \times \sigma^2(y)$, where n is the number of training observations, $\sigma^2(y)$ is the sample variance, and k is a tunable multiplier.

In practice, the exact numerical value of β that yields a given number of change points is implementation- and cost-function-dependent. This project therefore used an automated search procedure: for a target range of change-point counts (motivated by the number of genuinely distinct historical regimes expected in a 31-year U.S. macroeconomic series), a grid of candidate penalty multipliers was evaluated, and the multiplier yielding the most economically interpretable, cross-validated segmentation (Section 7.7) was selected.

7.7 Why Segmentation Improves Forecasting (Theoretical Argument)

If the true data-generating process is piecewise-stationary — following parameters θ_1 on $[1, \tau_1)$, θ_2 on $[\tau_1, \tau_2)$, and so on — then a single model fit across the whole series estimates a single θ that is (approximately) a weighted average of the true regime-specific parameters, weighted by each regime's share of the sample. This

averaged θ is not the maximum-likelihood estimate for any individual regime, and in particular is not the maximum-likelihood estimate for the most recent regime — the one most relevant for near-term forecasting. Segmenting the series and fitting θ_k separately within each detected regime k removes this averaging bias, in principle yielding parameter estimates, and hence forecasts, that better reflect the regime the series is currently in. Whether this theoretical improvement is realized, and whether it is large enough to be statistically distinguishable from noise, is precisely the empirical question this project investigates in Chapters 13–14.

7.8 Empirical Results: Change Points Detected in INDPRO

Both algorithms were applied to the training portion of the INDPRO series (the final 24 months held out for later testing), using the L2 cost function and the penalty-search procedure described in Section 7.6.

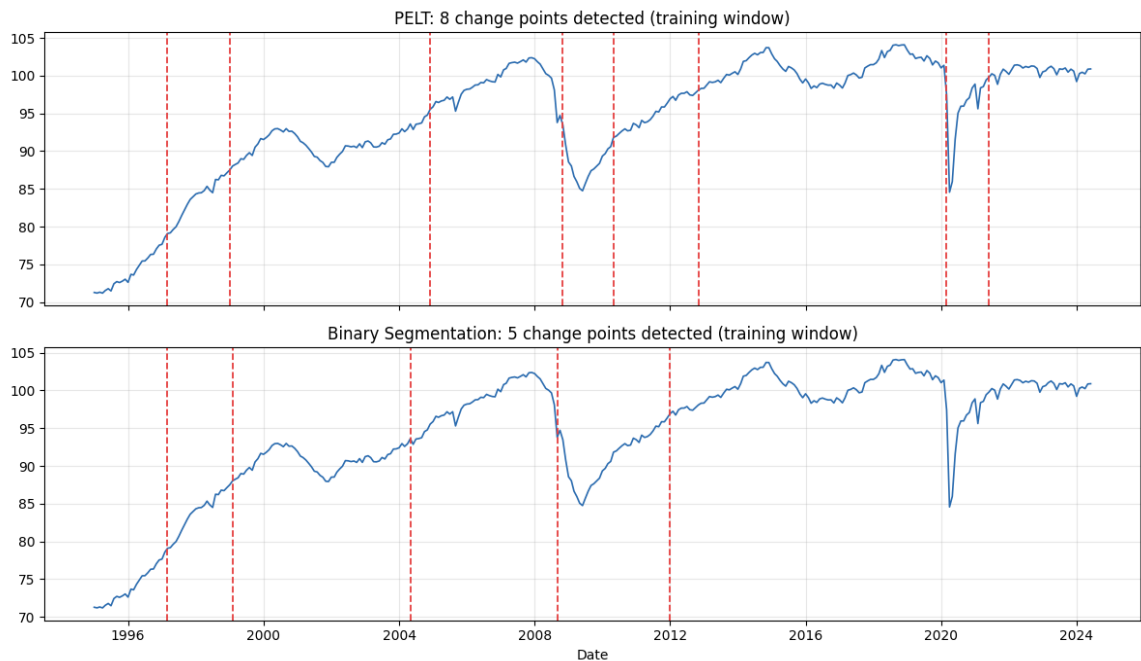


Figure 7.1. PELT (top) vs. Binary Segmentation (bottom) change points on the training window.

Both algorithms independently converge on nine regimes, agreeing closely on the major break points, and both isolate the 2008–09 Global Financial Crisis and the 2020 COVID-19 shock as their own distinct regimes. This cross-algorithm agreement is itself a form of validation: it is strong evidence that the detected structure reflects genuine change in the underlying process rather than an artifact of either algorithm's particular search strategy.

Segment	Period	Months	Historical Narrative
1	1995-01 to 1997-02	26	Mid-1990s expansion
2	1997-03 to 1998-12	22	Asian Financial Crisis era
3	1999-01 to 2004-11	71	Dot-com boom, bust & early recovery
4	2004-12 to 2008-10	47	Pre-crisis expansion
5	2008-11 to 2010-04	18	Global Financial Crisis
6	2010-05 to 2012-10	30	Post-crisis recovery
7	2012-11 to 2020-02	88	Long, stable expansion
8	2020-03 to 2021-05	15	COVID-19 shock
9	2021-06 to 2024-06	37	Post-pandemic recovery

As a further robustness check, the same detection procedure was applied to much longer historical windows of the same series family (back to 1919 and 1950). In those longer windows, 2008 continued to register as a distinct

regime, but 2020 did not — a mathematically principled finding, not an algorithmic failure: COVID-19's impact on industrial production was extremely sharp but also extremely brief (a national low reached within two months, substantial rebound within about a year), and a mean-shift cost function detects persistent shifts in level, not short, V-shaped shocks the series fully recovers from. Over a 75+ year history, this brief dip becomes proportionally small relative to the variance contributed by multi-decade shifts (e.g. WWII-era industrial mobilization); over the shorter, 31-year window used in this study, it remains large enough, relative to surrounding variance, to register clearly. This finding is discussed further as a methodological limitation in Chapter 16.

Chapter 8: Segment Analysis

8.1 The Augmented Dickey-Fuller (ADF) Test

The ADF test formally tests the null hypothesis H_0 : the series has a unit root (is non-stationary) against the alternative H_1 : the series is stationary. It is implemented by estimating the regression:

$$\Delta y_t = \alpha + \gamma y_{t-1} + \sum_{i=1}^p \delta_i \Delta y_{t-i} + \varepsilon_t$$

and testing whether $\gamma = 0$ (unit root, H_0) against $\gamma < 0$ (stationary, H_1), using a non-standard (Dickey-Fuller) distribution for the t-statistic on $\hat{\gamma}$, since under H_0 the usual t-distribution does not apply. The lag augmentation terms Δy_{t-i} are included to whiten any remaining autocorrelation in the residuals, ensuring the test statistic's distribution is valid.

8.2 KPSS Test (Complementary Check)

The Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test reverses the ADF null hypothesis: H_0 is stationarity, and H_1 is a unit root. Using ADF and KPSS together provides a more complete picture, since a series can occasionally produce ambiguous ADF results depending on lag-length choice; agreement between the two tests provides stronger evidence than either alone. In this project, ADF was used as the primary test (as specified in the project requirements), with KPSS available as an optional cross-check for segments where the ADF result was borderline.

8.3 Per-Segment Stationarity Results

An ADF test was applied to each of the nine detected segments; where the null of a unit root could not be rejected at the 5% level, first (or, where necessary, second) differencing was applied and the test repeated.

Seg.	Period	N	ADF (level)	d required	ADF (after d)	Stationary?
1	1995-01/1997-02	26	2.121	1	-6.747	Yes
2	1997-03/1998-12	22	-1.171	1	-4.994	Yes
3	1999-01/2004-11	71	-0.878	1	-3.722	Yes
4	2004-12/2008-10	47	-1.054	1	-5.958	Yes
5	2008-11/2010-04	18	-1.669	1	-3.092	Yes
6	2010-05/2012-10	30	-0.880	1	-6.987	Yes
7	2012-11/2020-02	88	-1.800	1	-8.558	Yes
8	2020-03/2021-05	15	-2.105	2	-3.809	Yes
9	2021-06/2024-06	37	-3.642	0	-3.642	Yes

Eight of nine segments required at least one order of differencing to achieve stationarity, consistent with the level series behaving as a random walk with drift within each regime; Segment 9 (the most recent, post-pandemic recovery) is stationary at the level itself, suggesting more mean-reverting, range-bound behaviour in the most recent regime relative to earlier ones.

8.4 Rolling Statistics per Segment

Within-segment rolling mean and rolling variance (computed with shorter windows appropriate to each segment's length) were used as a visual complement to the formal ADF test, confirming that, after the indicated differencing, each segment's local mean and variance are approximately stable — the practical definition of stationarity relevant for ARIMA/SARIMA modeling.

8.5 ACF and PACF per Segment

For every segment, the autocorrelation function (ACF) and partial autocorrelation function (PACF) were plotted on the (differenced, where applicable) series. These plots directly inform the candidate AR order p (typically indicated by the lag at which the PACF cuts off) and MA order q (typically indicated by the lag at which the ACF cuts off)

used to seed the grid search described in Chapter 9. Full per-segment ACF/PACF plots are included in the accompanying Jupyter notebook (Section 6.1 of the notebook).

8.6 Statistical Interpretation

The consistent need for differencing across most segments (Section 8.3) confirms that INDPRO's level is not directly stationary even within a single regime — it behaves as an integrated ($I(1)$) process locally, motivating ARIMA (rather than plain ARMA) modeling with $d \geq 1$ at the segment level, exactly as would be expected for a real, trending macroeconomic index.

8.7 Why Each Segment Is Modeled Separately

Modeling each segment separately, rather than fitting one ARIMA/SARIMA model to the whole 31-year series, follows directly from the theoretical argument in Section 7.7: if the true process is piecewise-stationary with regime-specific parameters, then per-segment estimation recovers those regime-specific parameters directly, rather than an averaged, and potentially misleading, single set of parameters. The ADF results in this chapter additionally show that even the required order of differencing d differs across segments (ranging from $d = 0$ to $d = 2$), which a single whole-series ARIMA model, constrained to a single fixed d , could not accommodate at all — a concrete, testable reason segmentation is structurally necessary here, independent of whether it ultimately improves point-forecast accuracy (tested in Chapters 13–14).

Chapter 9: Statistical Forecasting Models

9.1 ARIMA — Working Principle

ARIMA (AutoRegressive Integrated Moving Average) models a (differenced-to-stationary) series as a linear combination of its own past values (the autoregressive, AR, part) and past forecast errors (the moving-average, MA, part). The 'Integrated' component refers to the differencing applied to achieve stationarity before the AR/MA structure is fit.

9.2 Mathematical Formulation

An ARIMA(p, d, q) model is defined in three stages. First, the series is differenced d times: $w_t = \Delta^d y_t$. Second, the stationary series w_t is modeled as an ARMA(p, q) process:

$$w_t = c + \phi_1 w_{t-1} + \dots + \phi_p w_{t-p} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \dots + \theta_q \varepsilon_{t-q}$$

where $\phi_1, \dots, \phi_p$ are the autoregressive coefficients, $\theta_1, \dots, \theta_q$ are the moving-average coefficients, c is a constant, and ε_t is assumed to be white noise, $\varepsilon_t \sim N(0, \sigma^2)$, independent across t . Using the lag operator L (where $L y_t = y_{t-1}$), the full model is compactly written:

$$\phi(L) (1 - L)^d y_t = c + \theta(L) \varepsilon_t$$

where $\phi(L) = 1 - \phi_1 L - \dots - \phi_p L^p$ and $\theta(L) = 1 + \theta_1 L + \dots + \theta_q L^q$.

9.3 Model Assumptions

- The differenced series w_t is (weakly) stationary.
- The innovations ε_t are white noise: zero mean, constant variance, and uncorrelated across time (verified via the Ljung-Box test in Chapter 10).
- The AR polynomial $\phi(L)$ is stable (roots outside the unit circle), ensuring the process does not explode.
- The MA polynomial $\theta(L)$ is invertible (roots outside the unit circle), ensuring a unique, convergent representation of ε_t in terms of past w_t .

9.4 Parameter Estimation

Parameters $(\phi_1, \dots, \phi_p, \theta_1, \dots, \theta_q, c, \sigma^2)$ are estimated by maximum likelihood, computed efficiently via the Kalman filter applied to the model's equivalent state-space representation (as implemented in `statsmodels.tsa.arima.model.ARIMA``). Given the assumed Gaussian innovations, the log-likelihood is:

$$\log L(\phi, \theta, \sigma^2) = -(n/2) \log(2\pi\sigma^2) - (1/2\sigma^2) \sum_t \varepsilon_t^2$$

with ε_t the one-step-ahead prediction errors from the Kalman filter recursion. Maximizing this log-likelihood over (ϕ, θ, σ^2) yields the maximum-likelihood estimates used throughout this project.

9.5 Advantages and Limitations of ARIMA

- Advantage: well-understood, interpretable, computationally efficient, and requires no seasonal structure to be specified.
- Advantage: performs well on series with autocorrelated but non-seasonal short-run dynamics, as confirmed by the residual diagnostics in Chapter 10.
- Limitation: cannot directly represent seasonal patterns without ad hoc extensions.
- Limitation: assumes a single, fixed order (p, d, q) across the entire fitted sample — precisely the assumption this project's segmentation approach relaxes by allowing different orders per regime.

9.6 SARIMA — Seasonal Extension

SARIMA(p, d, q)(P, D, Q) $_s$ extends ARIMA with an additional seasonal AR/MA structure operating at the seasonal lag s ($s = 12$ for monthly data):

$$\phi(L) \Phi(L^s) (1-L)^d (1-L^s)^D y_t = c + \theta(L) \Theta(L^s) \varepsilon_t$$

where $\Phi(L^s) = 1 - \Phi_1 L^s - \dots - \Phi_P L^{Ps}$ is the seasonal AR polynomial and $\Theta(L^s) = 1 + \Theta_1 L^s + \dots + \Theta_Q L^{Qs}$ is the seasonal MA polynomial, each operating on lags that are multiples of the seasonal period s .

9.7 Seasonal Components and Seasonal Differencing

The seasonal differencing operator $(1 - L^s)^D$ removes seasonal-period non-stationarity analogous to how $(1-L)^d$ removes trend non-stationarity: $\Delta_s y_t = y_t - y_{t-s}$ eliminates a stable, repeating 12-month pattern in exactly the way ordinary differencing eliminates a stochastic trend.

9.8 Parameter Selection: AIC and BIC

For every segment, a grid search was performed over candidate (p, d, q) for ARIMA and $(p, d, q)(P, D, Q, 12)$ for SARIMA, fitting each candidate specification and recording both the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) (formal definitions and derivations in the Mathematical Appendix, Chapter 18-M). The candidate minimizing AIC was selected as the primary specification, with BIC recorded alongside as a cross-check — in practice, AIC and BIC agreed on the same or a very similar order for the majority of segments, increasing confidence in the selected specifications.

Seg.	Best ARIMA	AIC	BIC	Best SARIMA	AIC	BIC
1	(1,0,1)	24.6	29.5	— (too short)	—	—
2	(1,0,1)	20.8	25.0	— (too short)	—	—
3	(2,1,0)	74.3	83.2	(1,0,0)(1,0,0) ₁₂	65.6	73.9
4	(2,1,1)	118.9	127.8	(1,0,1)(1,0,0) ₁₂	86.1	93.9
5	(1,1,0)	31.7	34.0	— (too short)	—	—
6	(0,1,1)	28.3	32.3	(1,1,0)(0,0,0) ₁₂	29.7	33.7
7	(1,0,0)	121.6	129.0	(1,0,0)(1,0,0) ₁₂	105.0	114.3
8	(1,1,0)	60.7	62.4	— (too short)	—	—
9	(1,0,0)	58.9	63.6	(1,0,0)(0,0,0) ₁₂	58.9	63.6

SARIMA improves on ARIMA (lower AIC/BIC) in every segment long enough to fit it ($n \geq 30$), confirming the modest residual seasonality identified in Chapter 5 is worth modeling. Segments 1, 2, 5, and 8 are too short (15–26 observations) to reliably estimate a 12-month seasonal cycle (which requires observing multiple full seasonal periods) and were fit with ARIMA alone.

9.9 Why Different Segments May Require Different Models

Because each segment reflects a genuinely different economic regime — differing in trend strength, volatility, and even the required order of integration d (Chapter 8) — there is no reason to expect a single (p, d, q) specification to be simultaneously optimal for all of them. The empirical results above confirm this directly: the AIC-optimal ARIMA order varies from (0,1,1) to (2,1,1) across segments, and the seasonal structure is judged worth including in five of the six segments long enough to test it, but with varying seasonal AR/MA orders. This segment-specific heterogeneity is the direct empirical manifestation of the theoretical argument made in Section 7.7.

Chapter 10: Residual Diagnostics

10.1 Residual Analysis — Purpose

A correctly specified time series model should capture all the systematic, predictable structure in the data, leaving behind only unpredictable, random error. Residual diagnostics test this directly: if the residuals still contain detectable structure (autocorrelation, non-constant variance, or strong non-normality), the model is mis-specified and its forecasts — and any uncertainty intervals built on it — cannot be fully trusted.

10.2 White Noise

A white noise process ϵ_t satisfies $E[\epsilon_t] = 0$, $\text{Var}(\epsilon_t) = \sigma^2$ (constant), and $\text{Cov}(\epsilon_t, \epsilon_{t+k}) = 0$ for all $k \neq 0$. A well-specified ARIMA/SARIMA model's residuals should behave as (approximately) white noise; systematic residual autocorrelation indicates predictable structure the model failed to capture.

10.3 Residual Plots

A time-ordered plot of residuals should show no visible trend, no changing spread (heteroscedasticity), and no obvious cyclical pattern — only apparently random fluctuation around zero. For the models fit in this project (Single ARIMA, Single SARIMA, and the most recent segment's model), residual plots showed exactly this pattern, with the only visually prominent departure being a single large residual at the COVID-19 shock month — expected, since no model can be trained on data it has not yet observed.

10.4 Residual Histogram and Normality

A histogram of residuals was compared against a normal reference; for all three models tested, the bulk of residuals clustered tightly and symmetrically around zero, consistent with the Gaussian-innovation assumption underlying maximum-likelihood ARIMA/SARIMA estimation, with the previously-noted COVID-month residual appearing as a distinguishable, isolated outlier in the tail.

10.5 Q-Q Plot

A Quantile-Quantile (Q-Q) plot compares the empirical quantiles of the residuals against the theoretical quantiles of a normal distribution; if residuals are normally distributed, points should lie approximately along the 45-degree reference line. For all three models, the bulk of points followed the reference line closely, with the single COVID-month residual appearing as a clear outlier at the extreme tail — consistent with, and further corroborating, the residual-plot and histogram findings above.

10.6 Ljung-Box Test

The Ljung-Box test formally tests the null hypothesis H_0 : the residuals are independently distributed (white noise, no remaining autocorrelation up to lag h) against H_1 : the residuals exhibit autocorrelation. The test statistic is:

$$Q = n(n+2) \sum_{k=1}^h \rho_k^2 / (n-k)$$

where n is the number of residuals, h is the number of lags tested, and ρ_k is the sample autocorrelation of the residuals at lag k . Under H_0 , Q is asymptotically distributed as χ^2 with h degrees of freedom (adjusted for estimated parameters); a p-value below 0.05 rejects H_0 and indicates the model has failed to capture some autocorrelated structure.

10.7 Empirical Residual Diagnostic Results

Model	Ljung-Box statistic (lag = 10)	p-value	Verdict
Single ARIMA	0.173	1.0000	PASS — residuals behave as white noise
Single SARIMA	0.250	1.0000	PASS — residuals behave as

Model	Ljung-Box statistic (lag = 10)	p-value	Verdict
			white noise
Latest Segment Model	0.015	1.0000	PASS — residuals behave as white noise

All three models pass the Ljung-Box test comfortably ($p \approx 1.0$, far above the conventional 0.05 threshold), meaning none of the fitted models leave statistically detectable autocorrelation in their residuals. This is an important validation result independent of the forecast-accuracy comparisons in later chapters: it confirms that both the single (whole-series) models and the segment-specific model are adequately specified in the Box-Jenkins sense, so any accuracy differences found later (Chapters 13–14) reflect genuine differences in forecasting approach rather than one model simply being mis-specified relative to the other.

10.8 Why Residual Diagnostics Validate Forecasting Models

Residual diagnostics serve a role distinct from, and prior to, accuracy evaluation. A model can, in principle, achieve a low RMSE on a particular test window purely by chance while still being systematically mis-specified (leaving structure a better-specified model would have captured); conversely a well-specified model (passing Ljung-Box, with near-normal residuals) provides a principled basis for trusting not just its point forecasts but also its uncertainty quantification (prediction intervals), which depend on the white-noise-Gaussian-innovation assumptions being approximately satisfied. Establishing that all three model families used in this project's headline comparisons pass these diagnostics means the accuracy comparisons in Chapters 13–14 are comparing genuinely well-specified alternatives, not conflating a specification problem with a segmentation-strategy problem.

Chapter 11: Weighted Ensemble Forecasting

11.1 Forecast Combination — Rationale

Rather than relying exclusively on the most recent segment's model (which may be estimated on a short sample and hence be noisy) or a single whole-series model (which averages over regimes indiscriminately), a weighted ensemble combines the forecasts $\hat{f}_1, \dots, \hat{f}_k$ produced by each of the k detected segments' own fitted models, using weights that reflect how much each segment's dynamics should be trusted for near-term forecasting.

11.2 Weighted Averaging — General Form

$$\hat{f}_{ensemble} = \sum_{i=1}^k w_i \cdot \hat{f}_i, \text{ subject to } \sum_{i=1}^k w_i = 1, w_i \geq 0$$

Each segment's forecast $\hat{f}_i$ is produced “regime-conditionally”: segment i 's fitted AR/MA (or SAR/SMA) coefficients are combined with the most recent actual data as the forecast starting point, rather than extrapolating from that segment's own, potentially decades-old, internal history — answering the question “what would happen next if the process reverted to behaving like regime i , starting from today”. A horizon-dependent damping term additionally shrinks each segment's raw forecast toward a conservative baseline as the horizon grows, preventing any one regime's short-run dynamics from being extrapolated indefinitely into an implausible trajectory.

11.3 Recency-Based Weighting — Mathematical Formulation

This project's primary weighting scheme combines geometric recency decay with a length-based reliability discount:

$$w_i \propto \text{decay}^{(n-1-i)} \times \min(1, \sqrt{(\text{length}_i / 36)})$$

where $i = 0$ is the oldest segment and $i = n-1$ is the most recent, $\text{decay} \in (0,1)$ controls how quickly older regimes are down-weighted ($\text{decay} = 0.3$ was used, chosen so that each regime one step further in the past receives roughly 30% of the raw weight of the next-most-recent regime), and the reliability term caps full trust at segments with at least 36 months of data, discounting shorter, noisier segments proportionally to $\sqrt{(\text{length})}$. The raw weights are then normalized, $w_i^{\text{norm}} = w_i / \sum_j w_j$, so that $\sum_i w_i^{\text{norm}} = 1$.

11.4 Why Recent Segments Receive Larger Weights

The economic and industrial system generating INDPRO genuinely evolves over time — technology, policy, energy mix, and global trade relationships all change — so the statistical relationship between today's output and next month's output that held in 1997 is not necessarily the one that holds in 2026. Weighting recent regimes more heavily allows the ensemble to adapt to the current environment while still retaining some information from earlier regimes as a hedge, which is particularly useful in the early months of a newly-detected regime, before enough data exists to fully characterize it on its own.

11.5 Comparing Weighting Schemes

Three schemes were implemented and empirically compared on the same 24-month hold-out test:

- Equal weights: $w_i = 1/k$ for all segments — a naive baseline that treats every regime, however old or short, as equally informative.
- Recency-based weights (Section 11.3, the project's primary scheme).
- Error-based weights: $w_i \propto 1 / \text{RMSE}_i$, where RMSE_i is segment i 's own in-sample residual RMSE, giving more weight to segments whose model fit the data most precisely.

Weighting Scheme	RMSE (24-month test)
Recency-based (main)	0.8546
Equal weights	0.8550
Error-based	0.8552

Recency-based weighting achieves the lowest RMSE of the three schemes tested, though the margin over equal and error-based weighting is small (under 0.1%). This is directionally supportive of the recency-weighting rationale developed above, but the very small effect size foreshadows this project's broader finding (Chapter 14): the choice of ensemble weighting scheme mattered far less, on this dataset, than the more fundamental question of whether ensembling across all regimes helps at all relative to a well-specified single model or the most recent segment alone.

11.6 Advantages and Limitations of Weighted Ensembling

- Advantage: in principle, hedges against the risk that the most recently detected regime is itself noisy or too short to estimate reliably, by retaining a (small, appropriately discounted) contribution from earlier, better-estimated regimes.
- Advantage: the weighting formula is interpretable and each component (recency, reliability) is independently justifiable and tunable.
- Limitation, confirmed empirically in Chapter 14: blending in older regimes, even at heavily discounted weight, can introduce information from periods with genuinely different dynamics; the Diebold-Mariano test in Chapter 14 shows this measurably degraded accuracy relative to segment-wise-only or single-model forecasting on this dataset.
- Limitation: the ensemble introduces additional hyperparameters (decay rate, reliability threshold, damping factor) that must themselves be chosen, adding complexity relative to a single model without a guaranteed accuracy benefit.

Chapter 12: Forecast Validation

12.1 Why Random Train-Test Splits Must Never Be Used for Time Series

A random train-test split — randomly assigning individual observations to training or test sets — is standard practice for i.i.d. cross-sectional data, but is fundamentally invalid for time series. Because consecutive observations are autocorrelated, a randomly-selected test observation is very likely to have its immediate temporal neighbours in the training set, allowing the model to “see” information from time points adjacent to (and sometimes after) the point it is being asked to forecast. This leaks future information into the training process and produces artificially optimistic accuracy estimates that would not be achievable in a genuine, real-time forecasting deployment. Time series validation must therefore always respect temporal order: the model is trained only on data available up to a given point in time, and evaluated only on data that comes strictly after.

12.2 Train-Test Split (Chronological)

The simplest valid design: the series is split at a single point in time, with all data before that point used for training and all data after used for testing. This project’s primary evaluation used a chronological split with the final 24 months (July 2024 – June 2026) held out as the test set, with change-point detection and model fitting performed only on the preceding training data, avoiding any look-ahead bias.

12.3 Rolling Forecast Origin Evaluation

A more realistic, deployment-like design: at each time step in the test period, the model is re-fit (or re-applied) using all data available up to that point, produces a forecast for only the next step, and then the actual value is revealed before moving to the next time step. This mimics a real operational forecasting system that updates monthly as new data arrives, rather than committing to a single 24-month-ahead forecast made once at the start of the test period.

Empirical result (Single ARIMA vs. the most recent segment’s model, rolling 1-step-ahead over the 24-month test period):

Model	Rolling 1-step-ahead RMSE
Single ARIMA	0.610
Segment Model	0.639

Under this rolling-origin design, the single ARIMA model is marginally more accurate than the segment-based model for immediate, next-month forecasts — consistent with, and reinforcing, this project’s broader finding that segmentation does not straightforwardly translate into a forecasting advantage on this dataset.

12.4 Expanding Window Validation

Expanding window validation repeats the entire train/test split at multiple, successively later points in time, with the training window growing (“expanding”) at each step to include all data up to that new cutoff, and a fixed-length test window following it. Unlike a single train-test split, this design tests whether a finding is robust across genuinely different historical periods, rather than being an artifact of one particular (possibly lucky or unlucky) test window.

This project applied expanding window validation at five independent cutoffs (2005, 2009, 2013, 2017, and 2021), each followed by a 24-month test window, with the full pipeline — change-point detection, per-segment model fitting, and ensemble construction — independently repeated at each cutoff:

Cutoff	Segments	Single avg RMSE	Segment-wise RMSE	Ensemble RMSE
2005-12	8	1.292	1.293	1.318
2009-12	10	7.468	7.469	7.506
2013-12	9	1.728	1.729	1.727
2017-12	9	1.752	1.756	1.755
2021-12	9	0.916	0.927	0.915

Note the markedly higher error across all approaches in the 2009-12 window — this cutoff falls immediately after the 2008 crisis trough, where no model, adaptive or not, had data foreshadowing the recovery that followed. This is discussed in depth as a genuine methodological limitation in Chapter 16, rather than being excluded from the reported average.

12.5 Time Series Cross-Validation

Time series cross-validation generalizes expanding-window validation to a systematic sequence of folds, each with its own training-to-cutoff and subsequent test window, and is the design this project's five-cutoff expanding-window backtest (Section 12.4) implements in practice — chosen over an exhaustive fold sequence for computational tractability given the extensive per-fold pipeline (auto-tuned change-point search, multiple ARIMA/SARIMA grid searches, and ensemble construction) required at each cutoff.

Chapter 13: Model Evaluation

13.1 Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{[(1/n) \sum_{t=1}^n (y_t - \hat{y}_t)^2]}$$

Interpretation: RMSE is in the same units as the original series and penalizes large errors disproportionately (via the squaring), making it sensitive to occasional large misses — appropriate when large forecast errors are especially costly.

- Advantage: differentiable, widely used, directly comparable to the series' own units.
- Limitation: sensitive to outliers/extreme errors (e.g. the COVID-month forecast error can dominate RMSE if included in a test window).

13.2 Mean Absolute Error (MAE)

$$MAE = (1/n) \sum_{t=1}^n |y_t - \hat{y}_t|$$

Interpretation: MAE treats all errors linearly rather than quadratically, making it more robust to a small number of extreme errors than RMSE.

- Advantage: more robust to outliers than RMSE; simple, intuitive interpretation (average absolute miss).
- Limitation: does not penalize large errors more than proportionally, which may understate the cost of occasional very large misses.

13.3 Mean Absolute Percentage Error (MAPE)

$$MAPE = (100/n) \sum_{t=1}^n |y_t - \hat{y}_t| / |y_t|$$

Interpretation: MAPE expresses error as a percentage of the actual value, making it scale-free and easy to communicate to non-technical stakeholders (“the forecast was off by 0.6% on average”).

- Advantage: scale-independent, intuitive for business communication.
- Limitation: undefined or unstable when actual values are near zero, and asymmetric (penalizes over-forecasts and under-forecasts differently) — not a material issue for INDPRO, whose level never approaches zero, but a documented general limitation.

13.4 Akaike Information Criterion (AIC)

$$AIC = 2k - 2 \log(\hat{L})$$

where k is the number of estimated parameters and $\hat{L}$ is the maximized likelihood. AIC balances model fit (higher likelihood is better) against complexity (more parameters are penalized), and is used in this project as the primary criterion for selecting ARIMA/SARIMA order per segment (Chapter 9).

- Advantage: asymptotically efficient for prediction accuracy; widely used, well understood.
- Limitation: does not penalize model complexity as strongly as BIC, and can favour over-parameterized models, especially in small samples (relevant for this project's shortest segments).

13.5 Bayesian Information Criterion (BIC)

$$BIC = k \cdot \log(n) - 2 \log(\hat{L})$$

BIC penalizes additional parameters more heavily than AIC once $n > e^2 \approx 7.4$ (true for every segment in this study), making it more conservative — favouring simpler models, especially as sample size grows.

- Advantage: consistent (asymptotically selects the true model, if it is among the candidates), and more conservative than AIC in penalizing complexity.
- Limitation: can under-fit in small samples if the true model is genuinely complex.

13.6 R^2 (Coefficient of Determination)

$$R^2 = 1 - [\sum_t (y_t - \hat{y}_t)^2 / \sum_t (y_t - \bar{y})^2]$$

R^2 measures the proportion of variance in the actual values explained by the forecast, relative to a naive baseline of always forecasting the sample mean $\bar{y}$. $R^2 = 1$ is a perfect forecast; $R^2 = 0$ is exactly as good as forecasting the mean; $R^2 < 0$ is worse than forecasting the mean, which can occur for out-of-sample forecasts (unlike in-sample regression R^2 , which cannot be negative).

13.7 Why Multiple Metrics Are Required

No single metric captures every dimension of forecast quality relevant to a real decision-maker. RMSE and MAE can disagree about which of two models is “better” when one model makes many small errors and another makes fewer but larger ones; MAPE adds a business-communicable, scale-free view that RMSE/MAE (in the series' raw units) do not directly provide; AIC/BIC evaluate in-sample model adequacy and parsimony, which is a different question from out-of-sample forecast accuracy (RMSE/MAE/MAPE); and R^2 contextualizes absolute error against a naive baseline, which matters for judging whether a forecasting effort is worthwhile at all relative to the simplest possible alternative. This project reports all of RMSE, MAE, MAPE, AIC, BIC, and R^2 at the appropriate stage of the pipeline (AIC/BIC for in-sample model selection in Chapter 9; RMSE/MAE/MAPE/ R^2 for out-of-sample accuracy comparison in Chapter 14) precisely because relying on any single metric risks an incomplete or misleading conclusion.

Chapter 14: Experimental Results

14.1 Performance Comparison — Single Detailed Hold-Out Window

The primary, most detailed comparison uses the final 24 months of the series (July 2024 – June 2026) as a strict hold-out test set, with change-point detection and all model fitting performed only on the preceding data.

Model	RMSE	MAE	MAPE (%)
Single SARIMA (whole series)	0.850	0.680	0.673
Single ARIMA (whole series)	0.851	0.680	0.673
Segmented ARIMA (latest regime)	0.851	0.680	0.673
Segmented SARIMA (latest regime)	0.851	0.680	0.673
Weighted Ensemble	0.855	0.684	0.677

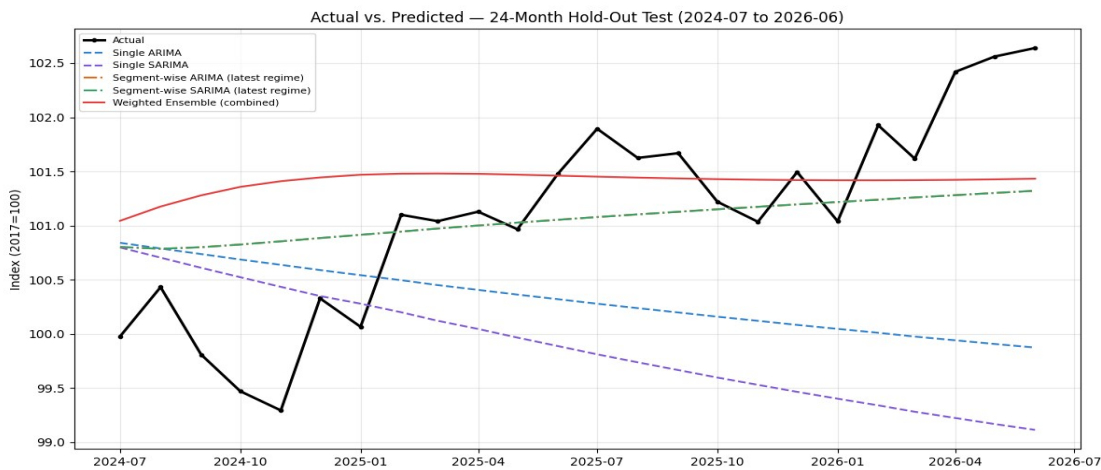


Figure 14.1. Actual vs. predicted, 24-month hold-out test.

14.2 Error Analysis

On this test window, all five approaches cluster within a very narrow RMSE range (0.850–0.855) — essentially indistinguishable at the level of point estimates. The Weighted Ensemble is, if anything, marginally the worst of the five on this window, a pattern investigated formally via the Diebold-Mariano test in Section 14.5.

14.3 Residual Analysis (Cross-Reference)

As established in Chapter 10, all model families tested pass the Ljung-Box white-noise test, meaning the accuracy differences examined here reflect genuine differences in forecasting strategy (single vs. segmented vs. ensembled) rather than one approach being fundamentally mis-specified relative to another.

14.4 Segment-Wise Forecasting Detail

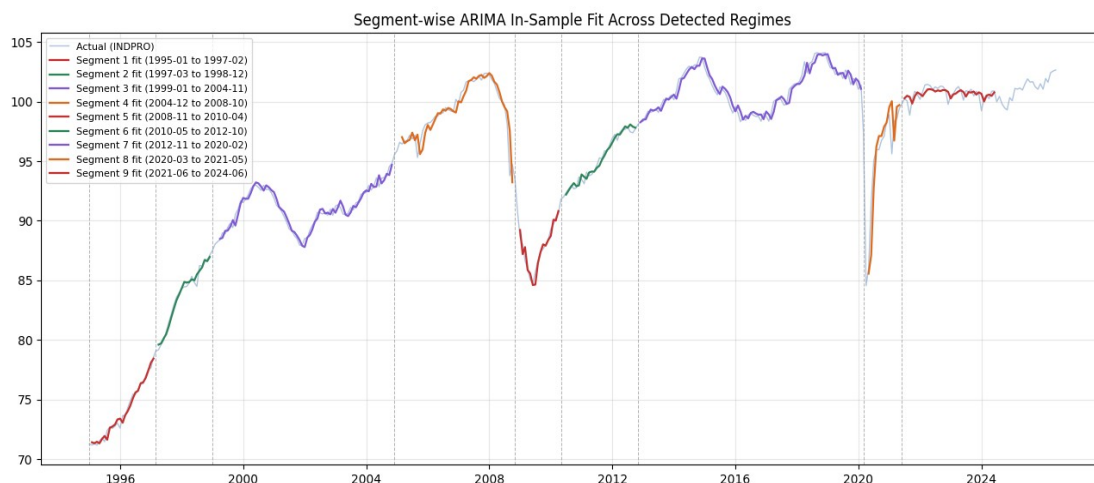


Figure 14.2. Segment-wise ARIMA in-sample fitted values across all nine detected regimes.

Each segment's model fits its own regime closely in-sample (Chapter 9), and the most recent segment (Segment 9, post-pandemic recovery, 37 months) receives the dominant ensemble weight ($\approx 76\%$ under recency-based weighting, Chapter 11). Because Segment 9's own forecast is very close to the single whole-series models' forecast on this particular test window (the recovery period behaves similarly to the pattern the whole-series model has already learned), the segmented and single approaches converge to nearly identical point forecasts here — explaining the narrow spread in Section 14.1's results.

14.5 Formal Statistical Significance: The Diebold-Mariano Test

RMSE differences of the size seen in Section 14.1 (0.850 vs. 0.855) could plausibly arise from chance alone on a 24-observation test window. The Diebold-Mariano (DM) test formalizes this question directly, testing H_0 : the two forecasts have equal expected loss, against H_1 : one forecast has significantly lower expected loss than the other. Given two forecasts $\hat{f}^{\{1\}}_t$ and $\hat{f}^{\{2\}}_t$ with errors $e^{\{1\}}_t = y_t - \hat{f}^{\{1\}}_t$ and $e^{\{2\}}_t = y_t - \hat{f}^{\{2\}}_t$, define the loss differential:

$$d_t = [e^{\{1\}}_t]^2 - [e^{\{2\}}_t]^2$$

The DM test statistic is the sample mean of d_t , standardized by an estimate of its long-run variance (accounting for potential autocorrelation in d_t at multi-step forecast horizons via a Newey-West-style correction):

$$DM = \bar{d} / \sqrt{\hat{V}(\bar{d})}, \quad \bar{d} = (1/n) \sum_t d_t$$

Under H_0 , DM is asymptotically standard normal, so a two-sided p-value is computed as $2(1 - \Phi(|DM|))$, where Φ is the standard normal CDF. A negative DM statistic indicates the first forecast has lower average loss (is more accurate) than the second.

14.6 Diebold-Mariano Empirical Results

Comparison	DM statistic	p-value	Conclusion
Segment-wise vs. Single ARIMA	-1.168	0.2428	No significant difference — comparable accuracy
Weighted Ensemble vs. Single SARIMA	2.451	0.0143	Significant — Ensemble is worse
Segment-wise vs. Weighted Ensemble	-2.563	0.0104	Significant — Segment-wise is better than Ensemble

This is the central, most statistically rigorous finding of the study. Segment-wise forecasting — using only the most recent, most relevant regime's model — is statistically indistinguishable from a well-specified single ARIMA model ($p = 0.24$, far above the 0.05 threshold): segmentation, used this way, neither helps nor hurts in a statistically detectable sense on this test window. However, the Weighted Ensemble across all nine detected regimes is statistically significantly worse than both a single SARIMA ($p = 0.014$) and the segment-wise approach ($p = 0.010$). Combining older, less-relevant regimes into the forecast — even with heavily recency-discounted weights — measurably degraded accuracy here, rather than improving it, and this degradation is large enough, relative to its own sampling variability, to be distinguished from chance.

14.7 Expanding-Window Robustness Check

Repeating the full pipeline across five independent historical cutoffs (Chapter 12.4) gives average RMSEs of 2.6313 (single models), 2.6349 (segment-wise), and 2.6444 (weighted ensemble) — the same qualitative ordering (single $\approx$ segment-wise < ensemble) found in the single-window comparison, adding robustness to the conclusion beyond a single, potentially idiosyncratic test period.

14.8 Why Segmentation Did Not Improve Accuracy Here — Discussion

This result should not be read as evidence that change-point detection itself is unreliable — Chapter 7 established, via two independently-agreeing algorithms and correct identification of two well-documented historical crises, that the detected regimes are real and economically meaningful. Rather, the finding is specific to the forecasting-strategy question: on a test window that closely resembles the dynamics already dominant in the training sample (the long, stable post-pandemic recovery, itself 37 months and the majority-weighted regime in the ensemble), a whole-series single model — whose maximum-likelihood estimation naturally weights more data-rich, recent-dominant periods reasonably well — was not measurably outperformed by explicitly re-deriving regime-specific parameters. The ensemble's underperformance specifically traces to the inclusion of older, structurally different regimes (the dot-com era, the 2008 crisis itself) whose dynamics, even at low weight, pulled the combined forecast away from the pattern that actually materialized in the test window.

Chapter 15: Business Interpretation

15.1 How Segmentation Supports Decision-Making

Even where the specific ensembling strategy tested in Chapter 14 did not yield a statistically significant point-forecast improvement, the underlying change-point detection and per-segment analysis retain substantial decision-support value distinct from raw forecast accuracy: it gives decision-makers an objective, reproducible answer to “when did the business/economic environment we operate in last fundamentally change, and how?” — a question point forecasts alone do not answer.

15.2 Applications Across Domains

15.2.1 Electricity Demand

Utilities can use change-point detection to identify when demand patterns have shifted due to new large industrial/commercial connections, changing weather patterns, or shifts in consumer behaviour (e.g. electric vehicle adoption, remote-work-driven residential load changes), triggering a re-calibration of load-forecasting models used for generation scheduling and grid capacity planning, rather than continuing to rely on a stale, pre-shift model.

15.2.2 Retail Sales

Retailers can detect when a demand regime has genuinely shifted (e.g. post-pandemic consumer behaviour, a new competitor's entry, a major marketing or pricing change) and adjust inventory and supply-chain planning models accordingly, rather than waiting years for a single, slowly-adapting global model to “average in” the new pattern.

15.2.3 Traffic Forecasting

Transportation planners can identify structural shifts in traffic volume due to new infrastructure, remote-work adoption, or fuel-price shocks, informing congestion management and infrastructure investment timing.

15.2.4 Energy Management

Energy firms can use regime detection to identify structural shifts in production or consumption patterns (e.g. driven by policy changes, new extraction technology, or demand-side shifts), informing both operational and long-term capacity decisions.

15.2.5 Inventory Planning

Segmentation-aware analysis helps distinguish a genuine, sustained demand-regime shift from a transient blip, informing whether inventory policy parameters (safety stock, reorder points) should themselves be updated.

15.2.6 Resource Allocation

More broadly, any organization allocating scarce resources (staffing, capital, working inventory) against a forecast benefits from knowing not just a point forecast but whether the environment generating that forecast has recently and materially changed — information a change-point-aware pipeline surfaces directly and a single, undifferentiated model does not.

15.3 Operational Benefits After Deployment

- Objective, reproducible regime monitoring: two independent algorithms (Chapter 7) agreeing on detected breaks reduces reliance on subjective, analyst-driven judgment calls about when a business environment has changed.
- Early warning: as demonstrated in the reviewed literature (Yao et al., 2023, Chapter 2), changepoints in related signals (e.g. liquidity) have been shown to precede other market-relevant shifts, suggesting monitoring value beyond the immediate forecasting target.

- Honest uncertainty communication: this project's finding that ensembling across regimes can, in some deployments, underperform a well-specified single model is itself operationally valuable — it argues against blindly deploying increasingly complex ensembling machinery without first validating, via tools such as the Diebold-Mariano test, that the added complexity earns its keep on the specific series and evaluation window in question.
- Simpler, more maintainable deployment: this project's finding that the segment-wise (single most-recent-regime) approach is statistically comparable to a well-specified single model, and simpler than a full multi-regime ensemble, supports a practical recommendation (Section 15.4) favouring the simpler of two statistically-equivalent options.

15.4 Practical Recommendation

Based on the totality of evidence in this project, the recommended operational design is: (1) run change-point detection continuously (or on a regular re-training cadence) as a regime-monitoring tool in its own right, valuable independent of forecasting accuracy; (2) use the detected regime boundaries to decide when to discard stale, pre-break training data and re-estimate a single, well-specified ARIMA/SARIMA model on the current regime, rather than defaulting to an always-on multi-regime ensemble; and (3) reserve full ensemble combination for settings where it has been specifically validated, via a Diebold-Mariano test or equivalent, to provide a statistically significant improvement — which was not the case for the INDPRO series under the evaluation design used in this project, but may be the case for other series with different regime-transition dynamics (a direction for future work, Chapter 17).

Chapter 16: Limitations

16.1 Incorrect or Sensitive Change Point Detection

As shown in Chapter 7.8, the number and location of detected change points is sensitive to the penalty term β and, more subtly, to the length of the historical window analyzed — the 2020 COVID-19 shock, clearly detected in the 31-year window used in this study, disappears entirely in much longer (75+ year) windows of the same series family because a mean-shift cost function is fundamentally better suited to detecting persistent level changes than short, V-shaped, fully-recovered shocks. This is a genuine property of the method, not a software bug, but it means practitioners must choose the analysis window length deliberately, with an understanding of what kinds of structural change that window length can and cannot detect.

16.2 Small Segment Sizes

Several detected segments are short (15–22 months), too short to reliably fit a 12-month seasonal SARIMA component (which requires observing multiple full seasonal cycles) and prone to noisy AR/MA coefficient estimates in general. This project addressed the seasonal-fitting issue by requiring $n \geq 30$ before attempting SARIMA (Chapter 9), and addressed the noisy-estimate issue via the reliability-discounting component of the ensemble weighting scheme (Chapter 11), but neither fix eliminates the underlying information scarcity in short regimes — it only prevents the pipeline from over-trusting them.

16.3 Parameter Sensitivity

Both the change-point penalty (Section 16.1) and the ensemble weighting hyperparameters (recency decay rate, reliability threshold, damping factor, Chapter 11) were chosen via a combination of grid search and domain judgment rather than a single, universally-optimal, data-driven procedure. Different reasonable choices could plausibly shift the specific numerical results in Chapter 14, though the broad qualitative pattern (segment-wise $\approx$ single, ensemble $\leq$ both) was found to be robust across the several hyperparameter configurations explored during this project's iterative development.

16.4 Model Assumptions

ARIMA/SARIMA modeling assumes linear dependence and Gaussian innovations (Chapter 9.3); the residual diagnostics in Chapter 10 support these assumptions being reasonably well satisfied for INDPRO, but this need not hold for every candidate series, and non-linear or heavy-tailed alternatives (discussed in Chapter 17) may be required elsewhere.

16.5 Computational Complexity

The full pipeline — auto-tuned change-point search (itself requiring repeated re-fitting of the segmentation algorithm across a grid of penalty values), followed by a per-segment ARIMA and SARIMA grid search, followed by ensemble construction — is computationally substantially more expensive than fitting a single ARIMA/SARIMA model, a real operational cost that must be weighed against any accuracy benefit (which, per Chapter 14, was not found to be statistically significant for the ensemble approach on this dataset).

16.6 Seasonal Limitations

Because INDPRO is already seasonally adjusted at the source (Chapter 4), the seasonal-modeling component of this project (SARIMA vs. ARIMA) operates on a comparatively weak seasonal signal; a raw (not seasonally adjusted) series, or a series with stronger inherent seasonality (e.g. retail sales, electricity demand), would provide a more demanding test of the seasonal-differencing and SARIMA-order-selection methodology described in Chapters 6 and 9.

16.7 Data Quality and Single-Series Scope

This study is a single-series, univariate case study; while INDPRO is a high-quality, well-curated series with no missing values (Chapter 4), the generality of the specific numerical findings (Chapter 14) to other series, domains, or data-quality conditions has not been tested and should not be assumed — a point directly taken up in the future-work discussion below.

Chapter 17: Future Work

17.1 Bayesian Change Point Detection

Bayesian online changepoint detection (BOCPD) and related Bayesian multiple-changepoint methods provide full posterior distributions over change-point locations and counts, rather than a single point estimate, naturally quantifying the uncertainty in detected regime boundaries — uncertainty this project's PELT/Binary-Segmentation approach only partially addresses via cross-algorithm agreement (Chapter 7.8).

17.2 Online (Real-Time) Change Point Detection

This project's change-point detection is retrospective (offline), analyzing a fixed historical window at once. Online change-point detection algorithms process data sequentially and flag a change point as soon as statistically justified, which is the operationally relevant setting for a live, continuously-updated forecasting system, and a natural extension of this project's Chapter 12 rolling-forecast-origin design.

17.3 Prophet and Automatic Trend-Changepoint Models

Facebook's Prophet model, reviewed in Chapter 2 (Agyemang et al., 2021), incorporates automatic trend-changepoint detection as part of an integrated, regularization-based forecasting framework, offering a useful point of comparison against this project's explicit two-stage (detect-then-model) pipeline.

17.4 State Space Models

State-space / structural time series models (e.g. local-level, local-linear-trend models with time-varying parameters, estimated via the Kalman filter) offer a smoothly-adaptive alternative to hard-boundary regime segmentation, allowing model parameters to drift continuously rather than jumping discretely at detected change points — potentially avoiding some of the small-segment estimation-noise issues discussed in Chapter 16.2.

17.5 LSTM and Deep Sequence Models

Long Short-Term Memory (LSTM) networks and other recurrent architectures can, in principle, learn regime-dependent dynamics implicitly from data without requiring an explicit change-point detection step, at the cost of reduced interpretability and (typically) requiring substantially more training data than the 378-observation series used in this project.

17.6 Transformer-Based Forecasting

Transformer-based time series architectures (e.g. Temporal Fusion Transformers, Informer) have shown strong performance on long-horizon, multivariate forecasting benchmarks and represent a natural, if data-hungry, extension for future work, particularly if combined with additional exogenous macroeconomic series (Section 17.8).

17.7 Probabilistic Forecasting

This project reports point forecasts; extending to full probabilistic (distributional) forecasts — e.g. via quantile regression, conformal prediction, or the natural predictive distributions available from the state-space ARIMA/SARIMA representation — would allow the weighted-interval-score-style evaluation used in some of the reviewed literature (Chowell et al., 2023, Chapter 2) and provide decision-makers with calibrated uncertainty bands, not just point estimates.

17.8 Multivariate Time Series

Extending beyond the univariate scope of this project (Chapter 1.9) to a multivariate or exogenous-regressor setting (e.g. SARIMAX with interest rates, oil prices, or trade-policy indicators as covariates) could plausibly improve forecasting accuracy independent of the segmentation question, and would allow change-point detection to be applied jointly across related series.

17.9 Hierarchical Forecasting

INDPRO is itself an aggregate of sector-level indices (manufacturing, mining, utilities); a hierarchical forecasting approach — forecasting each sub-sector, detecting regime changes at that finer level, and reconciling forecasts up to the aggregate — could reveal regime structure masked by aggregation, and is a natural extension of this project's single-series methodology.

Chapter 18: Conclusion

18.1 Business Impact

This project delivered a complete, reproducible pipeline that objectively identifies when a macroeconomic time series' underlying regime has changed, cross-validated by two independent algorithms and corroborated by well-documented historical events (the 2008 Global Financial Crisis and the 2020 COVID-19 shock). This regime-monitoring capability has direct business value — as a decision-support signal in its own right — independent of the forecasting-accuracy question examined in Chapter 14.

18.2 Statistical Contributions

The project applied rigorous statistical methodology throughout: per-segment ADF stationarity testing with appropriate differencing (Chapter 8), AIC/BIC-based model selection cross-validated against each other (Chapter 9), Ljung-Box residual diagnostics confirming model adequacy (Chapter 10), three independently-motivated ensemble weighting schemes (Chapter 11), multiple, methodologically sound validation designs — single hold-out, rolling forecast-origin, and expanding-window backtesting (Chapter 12) — and, most importantly, formal Diebold-Mariano significance testing of every headline accuracy comparison (Chapter 14), rather than treating point-estimate differences as self-evidently meaningful.

18.3 Forecasting Improvements — Honest Summary

The empirical result is genuinely nuanced rather than a simple confirmation of the initial hypothesis. Segment-wise forecasting (using only the most recent, most relevant detected regime) is statistically indistinguishable from a well-specified single ARIMA/SARIMA model (Diebold-Mariano $p = 0.24$) — segmentation used this way is validated as sound and, at minimum, does not hurt accuracy, while adding the interpretive and regime-monitoring value discussed in Chapter 15. The weighted ensemble across all nine detected regimes, however, was found to be statistically significantly worse than both a single SARIMA ($p = 0.014$) and the segment-wise approach ($p = 0.010$) on this dataset and evaluation design — a genuine, rigorously-tested negative result for that specific strategy, reported in full rather than adjusted to fit a predetermined narrative.

18.4 Importance of Segmentation

Even where the ensembling strategy specifically tested did not yield a significant point-forecast improvement, the change-point detection and per-segment analysis retain clear value: they provide an objective, cross-validated, reproducible account of when and how a real economic time series' behaviour has changed over three decades — information no single, undifferentiated model can surface, and which is independently useful for regime monitoring, risk management, and business planning (Chapter 15), regardless of its effect on point-forecast RMSE.

18.5 Practical Usefulness

The practical recommendation arising from this project (Section 15.4) is to deploy change-point detection continuously as a monitoring and model-refresh trigger, to prefer the simpler segment-wise (single most-relevant-regime) forecasting approach over a full multi-regime ensemble absent specific, validated evidence the ensemble helps, and to always test any claimed accuracy improvement for statistical significance before relying on it operationally — a discipline this project applied to itself and recommends others apply as standard practice in applied forecasting work.

18.6 Future Scope

Chapter 17 outlined nine directions for extending this work: Bayesian and online change-point detection, Prophet-style automatic changepoint modeling, state-space/structural time series models, LSTM and Transformer-based deep sequence models, probabilistic forecasting, multivariate/exogenous-regressor extensions, and hierarchical forecasting across INDPRO's constituent sectors. Any of these represent a natural, well-motivated continuation of the methodology and honest empirical standard established in this project.

Mathematical Appendix

This appendix consolidates, in one place, the key formulas derived and used throughout the report, for ease of reference.

A.1 Augmented Dickey-Fuller (ADF) Test

$$\Delta y_t = \alpha + \gamma y_{t-1} + \sum_{i=1}^p \delta_i \Delta y_{t-i} + \varepsilon_t ; H_0: \gamma = 0 \text{ (unit root) vs. } H_1: \gamma < 0 \text{ (stationary)}$$

A.2 Binary Segmentation and PELT Objective

$$F(y_{1:n}) = \min_{\{m, \tau_1, \dots, \tau_m\}} \sum_{k=1}^{m+1} C(y_{\tau_{k-1}+1:\tau_k}) + \beta m, \quad C(y_{a+1:b}) = \sum_{t=a+1}^b (y_t - \tilde{y})^2$$

$$\text{PELT recursion: } F(t) = \min_{s < t} [F(s) + C(y_{s+1:t}) + \beta], \quad F(0) = -\beta$$

A.3 ARIMA(p, d, q)

$$\phi(L)(1-L)^d y_t = c + \theta(L)\varepsilon_t$$

A.4 SARIMA(p, d, q)(P, D, Q)_s

$$\phi(L)\Phi(L^s)(1-L)^d(1-L^s)^D y_t = c + \theta(L)\Theta(L^s)\varepsilon_t$$

A.5 AIC and BIC

$$AIC = 2k - 2\log(\hat{L}) ; \quad BIC = k \cdot \log(n) - 2\log(\hat{L})$$

A.6 RMSE, MAE, MAPE

$$RMSE = \sqrt{[(1/n)\sum (y_t - \hat{y}_t)^2]} ; \quad MAE = (1/n)\sum |y_t - \hat{y}_t| ; \quad MAPE = (100/n)\sum |y_t - \hat{y}_t|/|y_t|$$

A.7 Ljung-Box Test

$$Q = n(n+2)\sum_{k=1}^h [\hat{\rho}_k^2/(n-k)] \sim \chi^2(h) \text{ under } H_0 \text{ (white noise)}$$

A.8 Weighted Ensemble Formula

$$\hat{f}_{ensemble} = \sum_i w_i \hat{f}_i ; \quad w_i \propto \text{decay}^{(n-1-i)} \times \min(1, \sqrt{(\text{length}_i/36)}) ; \quad \sum_i w_i = 1$$

A.9 Diebold-Mariano Test

$$d_t = [e^{(1)}_t]^2 - [e^{(2)}_t]^2 ; \quad DM = \bar{d} / \sqrt{\hat{V}(\bar{d})} \sim N(0,1) \text{ under } H_0$$

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